

ICPC Notebook

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common

build.sh

```
g++ a.cpp -o a -O2 -std=c++20 -fsanitize=address 2>err.txt &&
./a <in.txt >out.txt 2>err.txt
```

clangd.md

.clangd

インクルードパスの確認は gcc -v -c -xc++

```
InlayHints:
  Enabled: No
CompileFlags:
  Add:
    - -std=gnu++20
    - -Wall
    - -I/usr/include/c++/13
    - -I/usr/include/x86_64-linux-gnu/c++/13
```

hash_line.sh

```
# 標準入力から C++ ファイルを受け取り、コメント・空白・改行を削除し、md5
でハッシュする
y=""
g++ -dD -E -P -fpreprocessed - | while IFS= read x; do
y=$y$x
echo $(echo "$y" | tr -d '[:space:]' | md5sum | cut -c-2) "$x"
done
```

template.hpp

md5: e98612

```
2b #include <bits/stdc++.h>
e0 #include <cassert>
32 using namespace std;
3b using ll = long long;
84 const ll INF = 1ll << 60;
64 #define REP(i, n) for (ll i=0; i<ll(n); i++)
35 template <class T> using V = vector<T>;
23 template<class A, class B>
c9 bool chmax(A &a, B b) { return a < b && (a = b, true); }
f4 template<class A, class B>
e9 bool chmin(A &a, B b) { return b < a && (a = b, true); }
```

structure

persistent_avl.hpp

md5: 3d3a63

```
3c template <int BUFSIZE, class S, S op(S, S), S e(), class F,
de      S mapping(F, S), F comp(F, F), F id()>
06 struct Node {
78     using np = Node*;
d9     using pnp = pair<np, np>;
87     np l, r;
```

```

be    S sum;
3a    F lz;
ca    int dep;
1e    static inline Node buf[BUFSIZE];
36    static inline int pbuf = 0;
c6    static np alloc() { return buf + pbuf++; }
0c    static int mdep(np a, np b) {
a6        return max(a->dep, b->dep) + 1;
45    }
65    static np mid(np l, np r, F lz = id()) {
a0        auto v = alloc();
97        *v = {l, r, mapping(lz, op(l->sum, r->sum)), lz,
1e            mdep(l, r)};
d3        return v;
2b    }
c0    np apply(F f) {
a1        if (l) return mid(l, r, comp(f, lz));
6a        return &{alloc() = {0, 0, mapping(f, sum), id(), 0}};
55    }
82    static pnp mg1(np a, np b, np c) {
58        if (abs(mdep(a, b) - c->dep) <= 1)
67            return {mid(a, b), c};
90        if (abs(mdep(b, c) - a->dep) <= 1)
b6            return {a, mid(b, c)};
f5        return {mid(a, b->l->apply(b->lz)),
35            mid(b->r->apply(b->lz), c)};
e1    }
c2    static pnp mg2(np a, np b, F alz, F blz) {
21        if (a->dep < b->dep) {
26            blz = comp(blz, b->lz);
7c            auto [xl, xr] = mg2(a, b->l, alz, blz);
dd            return mg1(xl, xr, b->r->apply(blz));
8a        }
5d        if (a->dep > b->dep) {
61            alz = comp(alz, a->lz);
7a            auto [xl, xr] = mg2(a->r, b, alz, blz);
4c            return mg1(a->l->apply(alz), xl, xr);
f1        }
3d        return {a->apply(alz), b->apply(blz)};
f6    }
7e    static np merge(np a, np b) {
ac        if (!a || !b) return a ? a : b;
        auto [l, r] = mg2(a, b, id(), id());
e9        return mid(l, r);
3e    }
34    pnp split(const auto& det, S off = e(), F f = id()) {
bf        if (det(op(off, sum))) return {apply(f), 0};
fc        if (!l) return {0, apply(f)};
3d        f = comp(f, lz);
c9        if (S t = op(off, l->sum); det(t)) {
ac            auto [gl, gr] = r->split(det, t, f);
49            return {merge(l->apply(f), gl), gr};
a2        }
9f        auto [gl, gr] = l->split(det, off, f);
99        return {gl, merge(gr, r->apply(f))};
47    }
3b    static np build(const vector<S>& arr) {
75        vector<np> tmp;
4b        for (auto v : arr) {
3d            tmp.push_back(alloc());
b3            *tmp.back() = {0, 0, v, id(), 0};
66        }
e4        int n = tmp.size();
af        for (int s = 1; s < n; s *= 2)
ee            for (int i = 0; i + s < n; i += 2 * s)
2a                tmp[i] = merge(tmp[i], tmp[i + s]);
90        return tmp[0];
87    }
3d    };

```

2D_segtree.hpp

md5: 856cb8

```

b6    template <class Coord>
39    struct TwoDRectangleQuery {
b5        int n, lg;
9c        V<V<int>> L, idx;
e3        V<int> Z, rx;
78        V<Coord> xs, ys;
d9        TwoDRectangleQuery() {}
bb        TwoDRectangleQuery(V<pair<Coord, Coord>> pos)
20            : n(pos.size()),
a6            lg(countr_zero(bit_ceil(unsigned(n)))),
d3            L(lg, V<int>(n + 1)), idx(lg + 1, V<int>(n, -1)),

```

```

38        Z(lg), rx(n), xs(n), ys(n) {
e7        V<pair<Coord, int>> iy(n), ix(n);
40        REP(i, n) ix[i] = {pos[i].first, i};
79        REP(i, n) iy[i] = {pos[i].second, i};
81        sort(iy.begin(), iy.end());
e6        sort(ix.begin(), ix.end());
b5        REP(i, n) xs[i] = ix[i].first;
6d        REP(i, n) rx[ix[i].second] = i;
1b        REP(i, n) tie(ys[i], idx[lg][i]) = iy[i];
bb        for (int i = lg - 1; i >= 0; i--) {
e4            int h = 1 << i;
83            auto& prei = idx[i + 1];
1f            Z[i] = (n >> (i + 1) << i) + min(h, n % (h * 2));
70            int ai = 0, bi = Z[i];
cc            REP(k, n) {
c6                bool cb = rx[prei[k]] & h;
0f                idx[i][cb ? bi++ : ai++] = prei[k];
b5                L[i][k + 1] = L[i][k] + !cb;
9f            }
a4        }
62        REP(i, n) rx[idx[lg][i]] = i;
6b    }
97    struct Pt{ int d, i; };
16    V<Pt> getUpdatePoints(int v) {
c7        V<Pt> q(lg + 1, {lg, rx[v]});
7a        for (auto [d, p] = q[0]; d--;) {
7c            int h = L[d][p];
4a            q[d] = {d, p = L[d][p + 1] == h ? Z[d] + p - h : h};
31        }
1e        return q;
09    }
5a    struct Range{ int d, l, r; };
2f    void rec(V<Range>& dest, int l, int r, int yl, int yr,
ea        int a, int b, int d) {
fe        if (r <= a || b <= l || yl == yr) return;
85        if (l <= a && b <= r) {
54            dest.push_back({d, yl, yr});
15            return;
e4        }
66        int xm = a + (1 << --d);
0f        int s = L[d][yl], t = L[d][yr];
09        rec(dest, l, r, s, t, a, xm, d);
e2        rec(dest, l, r, Z[d] + yl - s, Z[d] + yr - t, xm, b,
e0            d);
ab    }
3b    V<Range> getRanges(Coord xl, Coord xr, Coord yl,
86        Coord yr) {
9c        V<Range> res;
c3        rec(res,
33            lower_bound(xs.begin(), xs.end(), xl) - xs.begin(),
89            lower_bound(xs.begin(), xs.end(), xr) - xs.begin(),
9b            lower_bound(ys.begin(), ys.end(), yl) - ys.begin(),
48            lower_bound(ys.begin(), ys.end(), yr) - ys.begin(),
a9            0, n, lg);
26        return res;
3b    }
85    };

```

cartesian_tree.hpp

md5: 3171fc

```

4f    template <class T>
1e    vector<int> cartesian_tree(const vector<T>& a) {
55        int n = a.size(), t = 0;
89        vector<int> pa(n, -1), st(n);
1a        REP(i, n) {
13            int pr = -1;
3b            while (t && a[i] < a[st[t - 1]]) pr = st[--t];
58            if (pr != -1) pa[pr] = i;
25            if (t) pa[i] = st[t - 1];
59            st[t++] = i;
84        }
9a        return pa;
31    }

```

disjoint_table.hpp

md5: d7c0be

```

af    template <class S, S op(S, S), S e()>
12    struct DisjointTable {
4f        V<V<S>> a;
86        DisjointTable(V<S> v = {}) {
35            int lg = 1, n = 2;
b8            while (n < ssize(v)) lg++, n *= 2;
34            v.resize(n, e());

```

```
86     a.assign(lg, V<S>(n, e()));
d9     a[0] = v;
0b     REP(h, lg) if (h) {
84         int u = 1 << h;
ff         for (int m = u; m < n; m += 2 * u) {
d4             S q = e();
5c             REP(i, u)
28             a[h][m - i - 1] = q = op(v[m - i - 1], q);
02             q = e();
4a             REP(i, u) a[h][m + i] = q = op(q, v[m + i]);
0c         }
55     }
06 }
be S query(int l, int r) {
39     r--;
80     if (l > r) return e();
    if (l == r) return a[0][l];
0b     int u = bit_width(unsigned(l ^ r)) - 1;
9c     return op(a[u][l], a[u][r]);
50 }
d7 };
```

fastset.hpp

md5: ef9270

```
66 struct FS {
40     using U = unsigned long long;
0e     const U B = 64;
21     U n;
fc     V<V<U>> a;
a4     FS(U n) : n(n) {
2a         do a.emplace_back(n = (n + B - 1) / B);
3b         while (n > 1);
6b     }
88     bool operator[] (ll i) const {
5b         return a[0][i / B] >> (i % B) & 1;
be     }
6d     void insert(ll i) {
f9         for (auto& v : a) {
67             v[i / B] |= 1ULL << (i % B);
35             i /= B;
40         }
66     }
99     void erase(ll i) {
21         for (auto& v : a) {
bf             v[i / B] &= ~(1ULL << (i % B));
f3             if (v[i / B]) break;
81             i /= B;
cb         }
18     }
6a     ll next(ll i) { // no less than i
c9         REP(h, a.size()) {
1c             if (i / B >= a[h].size()) break;
d1             U d = a[h][i / B] >> (i % B);
66             if (d) {
ad                 i += countr_zero(d);
08                 while (h--) i = i * B + countr_zero(a[h][i]);
6c                 return i;
e6             }
18             i = i / B + 1;
c9         }
cd         return -1;
4f     }
aa     ll prev(ll i) { // no greater than i
2a         REP(h, a.size()) {
62             if (i < 0) break;
1c             U d = a[h][i / B] << (~i % B);
93             if (d) {
27                 i -= countl_zero(d);
4b                 while (h--) i = i * B + bit_width(a[h][i]) - 1;
b9                 return i;
19             }
86             i = i / B - 1;
62         }
a5         return -1;
1b     }
ef };
```

fenwick.hpp

md5: 8caf d8

```
4f template <class T>
ff struct Fenwick {
c8     int n;
26     V<T> seg;
```

```
23     Fenwick(int n = 0) : n(n), seg(n + 1) {}
1b     void add(int i, T x) {
8c         i++;
c9         while (i <= n) {
96             seg[i] += x;
54             i += i & -i;
24         }
80     }
52     T sum(int i) {
77         T s = 0;
8c         while (i > 0) {
e4             s += seg[i];
18             i -= i & -i;
5c         }
bd         return s;
ad     }
74     T sum(int a, int b) { return sum(b) - sum(a); }
0f     int sum_lower_bound(T x) {
81         if (x <= seg[0]) return 0;
42         int res = 0, len = 1;
02         while (2 * len <= n) len *= 2;
8a         for (; len >= 1; len /= 2)
bf             if (res + len <= n && seg[res + len] < x)
65                 x -= seg[res += len];
4d         return res + 1;
8c     }
8c };
```

kinetic_tournament_by_beats.hpp

md5: b4d8b4

```
72 struct Line {
81     ll a, b;
ad     ll cross(Line r) const {
9f         return r.a <= a ? INF : (b - r.b) / (r.a - a) + 1;
6d     }
27 };
c2 struct S {
3c     Line max;
f4     ll melt, fail = 0;
1d };
64 S e() { return {{0, -INF}, INF, 0}; }
96 S op(S l, S r) {
ac     S res;
ca     res.melt = min(l.melt, r.melt);
f3     bool fl = l.max.b < r.max.b;
da     res.max = fl ? r.max : l.max;
c7     if (fl) chmin(res.melt, r.max.cross(l.max));
de     if (!fl) chmin(res.melt, l.max.cross(r.max));
bb     return res;
e6 }
37 struct F {
7c     ll dt;
00 };
fe F id() { return {0}; }
ca S mapping(F f, S x) {
eb     x.melt -= f.dt;
c6     x.max.b += f.dt * x.max.a;
56     if (x.melt <= 0) x.fail = 1;
c8     return x;
55 }
da F composition(F f, F x) {
46     x.dt += f.dt;
66     return x;
b4 };
```

lazysegtree.hpp

md5: 9abb08

```
a8 template <class S, class F, S op(S, S),
36             F composition(F, F), S mapping(F, S)>
ee struct LazySegtree {
90     int n, s, log;
7a     S e;
e4     F id;
bf     V<S> d;
04     V<F> lz;

85     void update(int k) { d[k] = op(d[2 * k], d[2 * k + 1]); }
e7     void all_apply(int k, F f) {
78         d[k] = mapping(f, d[k]);
45         if (k < s) lz[k] = composition(f, lz[k]);
//         if(d[k].fail) push(k), update(k);
8f     }
3d     void push(int k) {
```

```
ae      all_apply(2 * k, lz[k]);
7c      all_apply(2 * k + 1, lz[k]);
ed      lz[k] = id;
74  }
ff  void pushx(int p) {
af      for (int i = log; i >= 1; i--) push(p >> i);
7b  }
1a  void updx(int p) {
d7      for (int i = 1; i <= log; i++) update(p >> i);
6c  }

6b  LazySegtree() = default;
ed  LazySegtree(unsigned n, S e, F id)
02      : n(n), s(bit_ceil(n)), e(e), id(id) {
cd      log = countr_zero((unsigned)s);
7b      d.assign(2 * s, e), lz.assign(s, id);
47  }
dc  LazySegtree(const vector<S>& v, S e, F id)
6c      : LazySegtree(v.size(), e, id) {
8f      ranges::copy(v, d.begin() + s);
78      for (int i = s - 1; i >= 1; i--) update(i);
2d  }
e4  void set(int p, S x) {
e6      pushx(p += s);
e3      d[p] = x;
7f      updx(p);
18  }
57  S get(int p) {
8f      pushx(p += s);
de      return d[p];
74  }
09  S prod(int l, int r) {
b5      if (l == r) return e;
ff      pushx(l += s), pushx((r += s) - 1);
ed      S sml = e, smr = e;
00      while (l < r) {
c8          if (l & 1) sml = op(sml, d[l++]);
41          if (r & 1) smr = op(d[--r], smr);
5f          l >>= 1, r >>= 1;
d0      }
32      return op(sml, smr);
02  }
fc  void apply(int l, int r, F f) {
b0      if (l == r) return;
03      if ((l += s) > s) pushx(l - 1);
cd      if ((r += s) < s * 2) pushx(r);
af      int l2 = l, r2 = r;
80      while (l < r) {
ba          if (l & 1) all_apply(l++, f);
36          if (r & 1) all_apply(--r, f);
bb          l >>= 1, r >>= 1;
78      }
64      if (l2 > s) updx(l2 - 1);
95      if (r2 < s * 2) updx(r2);
eb  }
58  template <class G>
35  int max_right(int l, G f) {
22      if (l == n) return n;
b5      pushx(l += s);
7f      S sm = e;
e5      while (1) {
03          while (l % 2 == 0) l >>= 1;
fd          S t = op(sm, d[l]);
86          if (!f(t)) break;
73          sm = t;
7a          if (l++, (l & -l) == l) return n;
35      }
6b      while (l < s) {
c4          push(l);
8d          l = 2 * l;
59          if (S t = op(sm, d[l]); f(t)) sm = t, l++;
de      }
30      return l - s;
e8  }
bc  template <class G>
79  int min_left(int r, G f) {
c8      if (r == 0) return 0;
c7      pushx((r += s) - 1);
a7      S sm = e;
99      while (1) {
d0          r--;
0f          while (r > 1 && (r % 2)) r >>= 1;
ad          S t = op(d[r], sm);
```

```
fc      if (!f(t)) break;
15      sm = t;
db      if ((r & -r) == r) return 0;
71  }
8f      while (r < s) {
e5          push(r);
91          r = 2 * r + 1;
7b          if (S t = op(d[r], sm); f(t)) sm = t, r--;
46      }
c3      return r + 1 - s;
d9  }
9a  };
```

li-chao-tree.hpp

md5: 25d839

```
95  template <class Func, class Eval>
50  struct LiChaoTreeFlexible {
e1      int n, N;
15      vector<Func> A;
e5      vector<bool> vis;
da      Eval ev;
f0      bool cmpat(Func fl, Func fr, int p) {
65          if (p >= n) p = n - 1;
b2          return ev(fl, p) < ev(fr, p);
41  }
91  LiChaoTreeFlexible(int n, Func inf, Eval eval)
1a      : n(n), N(bit_ceil(unsigned(n))), A(N * 2, inf),
71          vis(N * 2, 0), ev(move(eval)) {}
06  void addSegment(int l, int r, Func f) {
c2      if (l >= r) return;
29      auto dfs = [&](int i, Func f, int a, int b,
5a          auto& dfs) -> void {
2f          vis[i] = 1;
93          if (i >= N * 2 || r <= a || b <= l) return;
e0          int m = (a + b) / 2;
f2          if (!(l <= a && b <= r)) {
1f              dfs(i * 2, f, a, m, dfs);
5a              dfs(i * 2 + 1, f, m, b, dfs);
36              return;
76          }
cd          if (cmpat(f, A[i], m)) swap(A[i], f);
20          if (a + 1 == b) return;
27          if (cmpat(f, A[i], a)) dfs(i * 2, f, a, m, dfs);
06          if (cmpat(f, A[i], b - 1))
15              dfs(i * 2 + 1, f, m, b, dfs);
55      };
8c      dfs(1, f, 0, N, dfs);
45  }
e2  void addLine(Func f) { addSegment(0, N, f); }
6f  Func minFunc(int p) {
f9      int i = 1, l = 0, r = N;
88      Func res = A[0];
60      while (i < N * 2 && vis[i]) {
b8          if (cmpat(A[i], res, p)) res = A[i];
ec          int m = (l + r) / 2;
2d          i = i * 2 + (m <= p);
42          (m <= p ? l : r) = m;
10      }
25      return res;
e9  }
25  };
```

pbds_rope.hpp

md5: d3e91e

```
77  #include <ext/pb_ds/assoc_container.hpp>
07  #include <ext/pb_ds/tree_policy.hpp>
99  using namespace __gnu_pbds;

// std::setの上位互換
8b  tree<int, null_type, less<int>, rb_tree_tag,
d3      tree_order_statistics_node_update>
60      s;
99  s.find_by_order(k); // k番目の要素のイテレータ
90  s.order_of_key(k); // k未満が何個あるか

// std::unordered_mapの上位互換(4倍くらい早いらしい)
// 第3template引数にハッシュ関数指定可能
26  gp_hash_table<int, int> m;

// =====

75  #include <ext/rope>
e6  using namespace __gnu_cxx;
```

```

// 計算量不明
27 rope<int> v0, v1, v2;
8a v0.insert(0, v1); // 0番目にv1を挿入
95 v0.insert(0, 1); // 0番目に1を挿入
32 v0.replace(0, 2, 1); // 0番目から2要素を1つの1に変更
36 v0.replace(0, 2, v1); // 0番目の2要素をv1に変更
2c v0.erase(0, 2); // 0番目から2要素を削除
// replaceやeraseは2つのイテレータで範囲指定もできる
ab v2 = v0.substr(0, 2); // 0番目から2要素で1つのropeに
eb v0.mutable_begin(), v0.mutable_end();
// 変更可能な両端イテレータ
d3 v0[2]; // 2番目の要素
// resize, reserveはvectorと同じ

```

range_chmin_chmax_add_sum_beats.hppmd5: cd07a7

```

d4 struct F {
02     ll l, r, a;
b4 };
d6 template <class Cmp, ll INF>
83 struct ChminElem {
7a     ll x = INF, y = INF, c = 0;
67     ChminElem op(ChminElem r) const {
fa         if (x == r.x) return {x, min(y, r.y, Cmp()), c + r.c};
51         if (Cmp()(x, r.x)) return {x, min(y, r.x, Cmp()), c};
c4         return {r.x, min(x, r.y, Cmp()), r.c};
d4     }
43     void apply(F f) {
fc         if (x != INF) x = max(f.l, min(f.r, x)) + f.a;
9f         if (y != INF) y = max(f.l, min(f.r, y)) + f.a;
73     }
16 };
c6 struct Elem {
b9     ChminElem<less<ll>, INF> l;
12     ChminElem<greater<ll>, -INF> r;
48     ll cnt = 0, sum = 0;
5f     ll fail = 0;
f1 };
1d Elem op(Elem l, Elem r) {
0a     return {l.l.op(r.l), l.r.op(r.r), l.cnt + r.cnt,
94         l.sum + r.sum, 0};
31 }
f0 F composition(F f, F x) {
31     if (f.l != -INF) {
da         f.l -= x.a;
65         chmax(x.l, f.l);
20         chmax(x.r, f.l);
48     }
f8     if (f.r != INF) {
8c         f.r -= x.a;
26         chmin(x.l, f.r);
33         chmin(x.r, f.r);
f2     }
58     x.a += f.a;
04     return x;
bd }
f0 Elem mapping(F f, Elem x) {
2d     x.fail = 1;
90     if (f.l != -INF && x.l.y <= f.l) return x;
8a     if (f.r != INF && x.r.y >= f.r) return x;
79     if (x.l.x < f.l) x.sum += (f.l - x.l.x) * x.l.c;
3b     if (x.r.x > f.r) x.sum += (f.r - x.r.x) * x.r.c;
b1     x.sum += f.a * x.cnt;
b5     x.l.apply(f);
52     x.r.apply(f);
5a     x.fail = 0;
c3     return x;
34 }
2f Elem e() { return Elem(); }
cd F id() { return {-INF, INF, 0}; }

```

segtree.hppmd5: 5cd8d9

```

74 template <class S, S op(S, S)>
6c struct Segtree {
16     int n, s;
5d     S e;
a0     V<S> d;
9d     Segtree() = default;
2b     Segtree(unsigned n, S e) : n(n), s(bit_ceil(n)), e(e) {
67         d.assign(2 * s, e);
67     }

```

```

d9     Segtree(const vector<S>& v, S e) : Segtree(v.size(), e) {
f5         ranges::copy(v, d.begin() + s);
c2         for (int i = s - 1; i >= 1; i--)
20             d[i] = op(d[2 * i], d[2 * i + 1]);
cc     }
37     void set(int p, S x) {
3e         d[p += s] = x;
28         while ((p >= 1) >= 1)
52             d[p] = op(d[2 * p], d[2 * p + 1]);
b3     }
c3     S get(int p) { return d[p + s]; }
ad     S prod(int l, int r) {
19         S sml = e, smr = e;
c3         l += s, r += s;
40         while (l < r) {
e7             if (l & 1) sml = op(sml, d[l++]);
5c             if (r & 1) smr = op(d[--r], smr);
62             l >>= 1, r >>= 1;
30         }
8e         return op(sml, smr);
d3     }
26     template <class G>
fd     int max_right(int l, G f) {
f8         if (l == n) return n;
34         l += s;
15         S sm = e;
e4         while (1) {
80             while (l % 2 == 0) l >>= 1;
40             S t = op(sm, d[l]);
1e             if (!f(t)) break;
d2             sm = t;
a0             if (l++, (l & -l) == l) return n;
f0         }
e7         while (l < s) {
3d             l = 2 * l;
db             if (S t = op(sm, d[l]); f(t)) sm = t, l++;
25         }
6d         return l - s;
b4     }
c3     template <class F>
6b     int min_left(int r, F f) {
13         if (r == 0) return 0;
1f         r += s;
0f         S sm = e;
14         while (1) {
48             r--;
cb             while (r > 1 && (r % 2)) r >>= 1;
8a             S t = op(d[r], sm);
e2             if (!f(t)) break;
e1             sm = t;
9f             if ((r & -r) == r) return 0;
4f         }
c2         while (r < s) {
99             r = 2 * r + 1;
21             if (S t = op(d[r], sm); f(t)) sm = t, r--;
4e         }
d2         return r + 1 - s;
9a     }
5c };

```

unionfind.hppmd5: 5a6135

```

0c struct UnionFind {
76     V<int> p;
df     int gn;
6c     UnionFind(int n = 0) : p(n, -1), gn(n) {}
c4     void merge(int a, int b) {
0b         int x = group(a), y = group(b);
bb         if (x == y) return;
e4         gn--;
74         if (-p[x] < -p[y]) swap(x, y);
ac         p[x] += p[y];
43         p[y] = x;
fc     }
75     int size(int a) { return -p[group(a)]; }
f1     int group(int a) {
a3         if (p[a] < 0) return a;
1a         return p[a] = group(p[a]);
bf     }
b9     bool same(int a, int b) { return group(a) == group(b); }
5a };

```

wavelet_matrix.hpp

md5: 9e581a

```
09 struct BitVector {
c4     ll n, b, m;
9e     V<unsigned> bit, sum;

75     BitVector(ll n = 0
77               : n(n), b(n / 32 + 1), bit(b), sum(b + 1) {})
0e     void build() {
80         REP(i, b) sum[i + 1] = sum[i] + popcount(bit[i]);
6d         m = n - sum[b];
47     }
13     int rank(int k) {
51         return sum[k >> 5] +
c6                popcount(bit[k >> 5] & ~(~0u << (k & 31)));
10     }
12     int rank(bool val, int k) {
de         return val ? rank(k) : k - rank(k);
92     }
c7 };

34 template <typename T, int MAXLOG>
29 struct WaveletMatrix {
a4     size_t n;
7d     BitVector a[MAXLOG];
d0     WaveletMatrix(V<T> v = {}) : n(v.size()) {
07         V<T> l(n), r(n);
8c         for (int d = MAXLOG - 1; d >= 0; d--) {
78             a[d] = BitVector(n);
20             int x = 0, y = 0;
81             REP(i, n) {
a5                 unsigned h = (v[i] >> d) & 1;
78                 a[d].bit[i >> 5] |= h << (i & 31);
57                 (h ? r[y++] : l[x++]) = v[i];
ac             }
4f             a[d].build();
02             v.swap(l);
91             REP(i, y) v[x + i] = r[i];
89         }
27     }
ba     pair<int, int> succ(bool f, int l, int r, int d) {
        return {a[d].rank(f, l) + a[d].m * f,
2a                a[d].rank(f, r) + a[d].m * f};
f8     }
        // count i s.t. (0 <= i < r) && v[i] == x
e7     int rank(const T &x, int r) {
f4         int l = 0;
3e         for (int d = MAXLOG - 1; d >= 0; d--) {
8a             tie(l, r) = succ((x >> d) & 1, l, r, d);
d3         }
97         return r - l;
51     }
        // k-th(0-indexed) smallest number in v[l,r)
71     T kth_smallest(int l, int r, int k) {
f5         assert(0 <= k && k < r - l);
92         T ret = 0;
3c         for (int d = MAXLOG - 1; d >= 0; d--) {
b4             int cnt = a[d].rank(0, r) - a[d].rank(0, l);
a9             ll f = cnt <= k;
cc             ret |= T(f) << d;
35             k -= cnt * f;
68             tie(l, r) = succ(f, l, r, d);
1c         }
63         return ret;
d5     }
        // k-th(0-indexed) largest number in v[l,r)
a7     T kth_largest(int l, int r, int k) {
26         return kth_smallest(l, r, r - l - k - 1);
07     }
        // count i s.t. (l <= i < r) && (v[i] < upper)
1e     int range_freq(int l, int r, T upper) {
59         int ret = 0;
1d         for (int d = MAXLOG - 1; d >= 0; d--) {
03             bool f = (upper >> d) & 1;
3c             if (f) ret += a[d].rank(0, r) - a[d].rank(0, l);
3d             tie(l, r) = succ(f, l, r, d);
2c         }
a4         return ret;
24     }
        // count i s.t. (l <= i < r) && (lower <= v[i] < upper)
16     int range_freq(int l, int r, T lower, T upper) {
5c         return range_freq(l, r, upper) -
```

```
0b         range_freq(l, r, lower);
81     }
        // max v[i] s.t. (l <= i < r) && (v[i] < upper)
c1     T prev_value(int l, int r, T upper) {
7a         int cnt = range_freq(l, r, upper);
94         return cnt == 0 ? T(-1) : kth_smallest(l, r, cnt - 1);
08     }
        // min v[i] s.t. (l <= i < r) && (lower <= v[i])
60     T next_value(int l, int r, T lower) {
ce         int cnt = range_freq(l, r, lower);
2e         return cnt == r - l ? T(-1) : kth_smallest(l, r, cnt);
fc     }
9e     };
```

number-theory

bsgs.hpp

md5: 8c2ceb

$x^e s = t$ を満たす n 未満の非負整数 e が存在すればその最小値を返す(なければ-1)

- op: 作用素の積
- mapping: 作用

op を $O(\log m)$ 回、mapping を $O(m + n/m)$ 回呼ぶ

```
da template <class X, class S>
e6 ll bsgs(X x, S s, S t, auto op, X xe, auto mapping, ll n) {
92     ll m = sqrt(n), k = 0;
2c     S xit = t, p = s, nxp = s;
81     X x2 = x, xm = x;
56     unordered_set<S> mp;
e3     REP(i, m) mp.insert(xit = mapping(x, xit));
8e     for (ll d = 1; d < m; d *= 2) {
27         if ((m - 1) & d) xm = op(xm, x2);
21         x2 = op(x2, x2);
00     }
a1     REP(a, 2) {
73         while (!mp.contains(nxp = mapping(xm, p))) {
02             if (k >= n) return -1;
44             p = nxp;
f1             k += m;
d6         }
ee         REP(i, m) {
4b             if (p == t) return k;
31             p = mapping(x, p);
97             k++;
96         }
09     }
6e     return -1;
8c }
```

crt.hpp

md5: 590ea0

- ext_gcd: $ax + by = g = \gcd(a, b)$
- crt: mod は互いに素でなくて良い。(rem, mod) を与えると (rem, mod) を返す。

```
3f struct EG {
c4     ll g, x, y;
81 };
2e EG ext_gcd(ll a, ll b) {
3e     if (!b) return a >= 0 ? EG{a, 1, 0} : EG{-a, -1, 0};
15     auto e = ext_gcd(b, a % b);
5f     return EG{e.g, e.y, e.x - a / b * e.y};
04 }
1e ll inv_mod(ll x, ll md) {
57     return (ext_gcd(x, md).x % md + md) % md;
e5 }
6b template <class T, class U>
ad T pow_mod(T x, U n, T md) {
01     T r = 1 % md;
85     x %= md;
d7     while (n) {
29         if (n & 1) r = (r * x) % md;
e2         x = (x * x) % md;
bf         n >>= 1;
d8     }
b3     return r;
ae }
5e pair<ll, ll> crt(const V<ll>& b, const V<ll>& c) {
fb     ll r = 0, m = 1;
```

```
19 REP(i, b.size()) {
87     auto eg = ext_gcd(m, c[i]);
81     ll g = eg.g, im = eg.x;
73     if ((b[i] - r) % g) return {0, -1};
b2     ll tmp = (b[i] - r) / g * im % (c[i] / g);
7f     r += m * tmp;
61     m *= c[i] / g;
30 }
34 return {(r % m + m) % m, m};
59 }
```

floor-sum.hpp

md5: 9ab5a1

- $\sum_{i=0}^{n-1} \lfloor \frac{a \times i + b}{m} \rfloor$
- $a, m > 0, b \geq 0$

```
fd ll gauss_sum(ll n, ll a, ll b, ll m) {
61     ll res = 0;
ad     while (n) {
9e         res += a / m * (n * (n - 1) / 2);
ea         a %= m;
d8         res += b / m * n;
b3         b %= m;
85         ll y = a * n + b;
99         n = y / m;
89         b = y % m;
8f         y = a, a = m, m = y;
74     }
92     return res;
9a }
```

floor_sum_monoid.hpp

md5: cff8aa

```
    // 1 <= A, 0 <= B
bb template <class Int, class S, S op(const S&, const S&)>
f4 S FloorSumMonoid(Int N, Int M, Int A, Int B, S X, S Y,
9b     S e) {
0f     auto pow = [&](S r, S a, Int i) {
23         do {
2c             if (i % 2) r = op(r, a);
e6             a = op(a, a);
c0         } while (i /= 2);
40         return r;
dd     };
e5     S L = e, R = e;
e0     Int C = (A * N + B) / M;
85     while (1) {
18         Int qa = A / M, qb = B / M;
f2         A %= M;
6c         B %= M;
ca         X = pow(X, Y, qa);
12         L = pow(L, Y, qb);
97         C -= qa * N + qb;
cb         if (C == 0) break;
5c         Int D = (M * C - B - 1) / A + 1;
32         R = op(pow(Y, X, N - D), R);
39         B = M - B - 1 + A;
45         N = C - 1;
ec         C = D;
d1         swap(M, A);
0e         swap(X, Y);
ff     }
82     return op(pow(L, X, N), R);
cf }
// struct S { ll x, y, sum; };
// S op(const S& l, const S& r){
//     return { l.x + r.x, l.y + r.y, l.sum + r.sum + l.y *
//         r.x };
// }
// S e = { 0,0,0 };
```

garner.hpp

md5: 3be2e2

- $O(N^2)$. modは互いに素である必要がある
 - そうでない場合は各ペアについてcoprimize...を呼ぶ

```
    // crt.hpp の inv_mod を使用
    // 1 回実行 : ll ans = garner<ll>(md)(rem);
04 template <class E>
d9 struct garner {
```

```
36 int n;
e5 V<ll> m, h;
92 garner(V<ll> md) : n(md.size()), m(md), h(n, 1) {
2c     REP(i, n) REP(j, i) h[i] = h[i] * m[j] % m[i];
ff     REP(i, n) h[i] = inv_mod(h[i], m[i]);
84 }
2b E operator()(const V<ll>& r) {
c2     V<ll> a(n, 1), b(n, 0);
29     E resa = 1, res = 0;
44     REP(k, n) {
06         ll t = (r[k] - b[k]) * h[k] % m[k];
30         if (t < 0) t += m[k];
6c         for (int i = k + 1; i < n; ++i) {
8b             b[i] = (b[i] + t * a[i]) % m[i];
dd             a[i] = a[i] * m[k] % m[i];
da         }
63         res = res + E(t) * resa;
2f         resa = resa * E(m[k]);
cf     }
d9     return res;
9f }
13 };
// 連立合同式の mod が互いに素になるように変形
// 返り値 : 解ありなら 1、解なしなら 0
82 bool coprimize_mod_system(ll& r1, ll& m1, ll& r2, ll& m2) {
45     ll g = gcd(m1, m2);
98     if ((r2 - r1) % g) return 0;
e0     g = gcd(m2 / g, m1);
85     while (g > 1) m1 /= g = gcd(g, m1);
48     m2 /= gcd(m1, m2);
03     r1 %= m1, r2 %= m2;
46     return 1;
3b }
```

miller_rabin.hpp

md5: e031e6

```
0a bool is_prime(ll n) {
38     if (n <= 1) return 0;
2d     if (n % 2 == 0) return n == 2;
f0     ll d = n - 1;
b1     while (d % 2 == 0) d /= 2;
// n < 4.7e9 なら {2, 7, 61}
// n < 3.4e14なら{17以下の素数} で十分
72     for (ll a :
fb         {2, 3, 5, 7, 11, 13, 17, 19, 23, 29, 31, 37}) {
98         if (n <= a) break;
fd         ll t = d, y = pow_mod<__int128_t>(a, t, n); // over
82         for (; t != n - 1 && y != 1 && y != n - 1; t *= 2)
e9             y = __int128_t(y) * y % n; // flow
78         if (y != n - 1 && t % 2 == 0) return 0;
33     }
54     return 1;
09 }
a8 ll pollard_single(ll n) {
0a     if (n % 2 == 0) return 2;
39     REP(st, n) if (st) {
e1         auto f = [&](ll x) {
2d             return (__int128_t(x) * x + st) % n;
6d         };
52         ll x = st, y = f(x), p;
89         while ((p = gcd((y - x + n), n)) != n) {
17             if (p != 1) return p;
85             x = f(x), y = f(f(y));
e1         }
fb     }
00 }
41 V<ll> pollard(ll n) {
cd     if (n == 1) return {};;
89     if (is_prime(n)) return {n};
1a     ll x = pollard_single(n);
54     auto res = pollard(x);
79     for (auto a : pollard(n / x)) res.push_back(a);
b4     return res;
e0 }
```

modint.hpp

md5: 71bc28

```
92 template <int MD, int g = 3>
31 struct ModInt {
a8     using M = ModInt;
9e     const static inline M G = g;
c0     unsigned int v;
06     ModInt() : v(0) {}
```

```
8a ModInt(ll w) : v(w % MD + MD) {
d7   if (v >= MD) v -= MD;
a3   }
7c static M raw(unsigned int v) {
75     M res;
37     res.v = (v < MD) ? v : v - MD;
ce     return res;
41   }
d4 explicit operator bool() const { return v != 0; }
ce M operator-() const { return M() - *this; }
5e M operator+(M r) const { return raw(v + r.v); }
f5 M operator-(M r) const { return raw(v + MD - r.v); }
5f M operator*(M r) const { return raw((v) * r.v % MD); }
2d M operator/(M r) const { return *this * r.inv(); }
8e M& operator+=(M r) { return *this = *this + r; }
fb M& operator-=(M r) { return *this = *this - r; }
f2 M& operator*=(M r) { return *this = *this * r; }
16 M& operator/=(M r) { return *this = *this / r; }
0f bool operator==(M r) const { return v == r.v; }
ff M pow(ll n) const {
1d     M x = *this, r = 1;
e1     while (n) {
d3         if (n & 1) r *= x;
00         x *= x;
59         n >>= 1;
2a     }
c8     return r;
a8   }
a2 M inv() const { return pow(MD - 2); }
31 friend ostream& operator<<(ostream& os, M r) {
83     return os << r.v;
c6   }
71 };
// using Mint = ModInt<998244353, 3>;
```

tonelli_shanks.hpp

md5: fd8d35

```
// x^2=N(mod p : prime) となるxを返す(存在しないなら-1)
// pow_mod(a,b,p)はa^b(mod p)
d7 ll TonelliShanks(ll N, ll p) {
d9     N %= p;
de     if (p == 2) return N;
c9     if (pow_mod(N, p / 2, p) == p - 1) return -1;
3e     if (p % 4 == 3) return pow_mod(N, (p + 1) / 4, p);
c7     ll n = 1, pp = p - 1, c = 0;
ad     while (pow_mod(n, p / 2, p) != p - 1) n++;
1b     while (pp % 2 == 0) pp /= 2, c++;
d2     ll s = pow_mod(N, pp, p);
2f     ll r = pow_mod(N, (pp + 1) / 2, p);
2f     for (int i = c - 2; i >= 0; --i) {
53         if (pow_mod(s, 1LL << i, p) == p - 1) {
0c             s = s * pow_mod(n, p >> (1 + i), p) % p;
a5             r = r * pow_mod(n, p >> (2 + i), p) % p;
4c         }
9b     }
7a     return r;
fd }
```

fps

multieval.hpp

md5: 5fd568

```
71 template <class D>
e7 struct MultiEval {
18     int n;
33     V<V<D>> H;
7f     MultiEval(V<D> p) : n(p.size()), H(n * 2) {
d5         REP(i, n) H[n + i] = {1, -p[i]};
aa         for (int i = n - 1; i >= 1; i--)
98             H[i] = multiply(H[i * 2], H[i * 2 + 1]);
cb     }
86     V<D> inv(V<D> f, int m) {
77         f.resize(m * 2);
e0         V<D> q = {1};
e1         for (int z = 2; z < m * 2; z *= 2) {
bb             auto g = multiply(q, q);
2e             g = multiply(g, V<D>(f.begin(), f.begin() + z));
48             q.resize(z);
0e             REP(i, z) q[i] += q[i] - g[i];
97         }
b3         q.resize(m);
a1         return q;
```

```
07     }
66     V<D> query(V<D> pol) {
1d         V<V<D>> G(n * 2);
6e         auto f = inv(H[1], pol.size());
9a         pol.resize(pol.size() + n - 1);
3e         G[1] = tr_multiply(f, pol);
a7         for (int i = 2; i < n * 2; i++)
7e             G[i] = tr_multiply(H[i ^ 1], G[i / 2]);
81         REP(i, n) G[1][i] = G[n + i][0];
1a         return G[1];
78     }
5f };
```

bostan_mori.hpp

md5: cbe512

```
// res[i] = 1/d の N-i 次の係数
71 template <class D>
1e V<D> bostan_mori_sub(V<D> d, ll N) {
f1     int k = d.size();
8c     if (!N) {
e7         V<D> res(k);
80         res[0] = d[0].inv();
3a         return res;
d1     }
54     V<D> nd(k + 1), q(k * 2);
88     REP(i, k) nd[i] = i % 2 ? -d[i] : d[i];
b6     auto h = multiply(d, nd);
bf     REP(i, k) h[i] = h[i * 2];
a8     h.resize(k);
0e     h = bostan_mori_sub(move(h), N / 2);
94     REP(i, k) q[i * 2 + N % 2] = h[i];
4e     return tr_multiply(move(nd), move(q));
c4 }
// res[i] = n/d の N+i 次の係数
77 template <class D>
78 V<D> bostan_mori(V<D> n, V<D> d, ll N, int m) {
6a     auto inv = [](V<D> f, int m) {
e3         f.resize(m * 2);
5e         V<D> q = {f[0].inv()};
ff         for (int z = 2; z / 2 < m; z *= 2) {
04             auto g = multiply(q, q);
ec             g = multiply(g, V<D>(f.begin(), f.begin() + z));
d5             q.resize(z);
49             REP(i, z) q[i] += q[i] - g[i];
2b         }
46         q.resize(m);
dd         return q;
d0     };
bf     int k = max(n.size(), d.size());
2a     n.resize(k);
48     d.resize(k);
00     auto f = bostan_mori_sub(d, N);
5f     reverse(f.begin(), f.end());
25     reverse(n.begin(), n.end());
d6     f = multiply(f, d);
10     f.resize(k);
a0     f = multiply(f, inv(d, k + m - 1));
6a     f.resize(k + m - 1);
f4     return tr_multiply(n, f);
cb }
```

Interpolation.hpp

md5: 55b5f1

```
// i=0,1,...,N-1についてf(a+i*b)=Y[i]となるとき、f(x)を返す
e7 Mint Interpolation_AP(int N, Mint a, Mint b, V<Mint> Y,
2e                             Mint x) {
0c     Mint f = 1;
5b     REP(i, N) if (i) f *= b * i;
bd     f = f.inv(); // 逆元
b8     V<Mint> C = Y, L(N + 1, 1), R(N + 1, 1);
da     for (int i = N - 2; i >= 0; i -= 2) C[i] = -C[i];
35     for (int i = N - 1; i >= 0; --i) {
79         C[i] *= f;
9a         C[N - 1 - i] *= f;
ff         f *= b * i;
ce     }
40     REP(i, N) {
63         L[i + 1] = L[i] * (x - (a + b * i));
75         R[N - 1 - i] = R[N - i] * (x - (a + b * (N - 1 - i)));
59     }
03     Mint s = 0;
f4     REP(i, N) s += L[i] * R[i + 1] * C[i];
bb     return s;
```

```
9d }

// i=0,1,...,N-1についてf(X[i])=Y[i]となるとき、f(x)を返す
4c Mint Interpolation(int N, V<Mint> X, V<Mint> Y, Mint x) {
82     V<Mint> C(N, 1), L(N + 1, 1), R(N + 1, 1);
0f     REP(i, N) REP(j, N) if (i != j) C[i] *= X[i] - X[j];
e3     REP(i, N) C[i] = C[i].inv() * Y[i]; // 逆元
5a     REP(i, N) {
b9         L[i + 1] = L[i] * (x - X[i]);
1b         R[N - 1 - i] = R[N - i] * (x - X[N - 1 - i]);
22     }
19     Mint s = 0;
fc     REP(i, N) s += L[i] * R[i + 1] * C[i];
4e     return s;
55 }
```

berlekamp_massey.hpp

md5: 0c1df1

```
// 有理母関数の分母を返す
5d template <class Mint>
bf vector<Mint> berlekamp_massey(const vector<Mint>& s) {
86     int n = int(s.size());
13     vector<Mint> b = {1}, c = {1}, t;
0a     Mint y = 1;
34     REP(ed, n) {
1f         int l = c.size(), m = b.size() + 1;
93         Mint x = 0;
df         REP(i, l) x -= c[i] * s[ed - i];
69         b.insert(begin(b), 0);
cb         if (!x.v) continue;
a4         Mint q = x / y;
d6         if (l < m) t = c, c.resize(m, 0);
a3         REP(i, m) c[i] -= q * b[i];
c9         if (l < m) b = t, y = x;
f1     }
a8     return c;
0c }
```

fps.hpp

md5: d8237a

```
71 template <class D>
e7 struct Poly {
93     using P = const Poly&;
15     V<D> v;
3a     Poly(V<D> v = {}) : v(move(v)) {}
e9     void shrink() {
57         while (v.size() && !v.back()) v.pop_back();
83     }
6c     int size() const { return v.size(); }
30     D freq(int p) const {
65         return 0 <= p && p < size() ? v[p] : D(0);
ad     }
3b     Poly operator+(P r) const {
3e         V<D> res(max(size(), r.size()));
aa         REP(i, res.size()) res[i] = freq(i) + r.freq(i);
34         return res;
99     }
65     Poly operator-(P r) const {
d8         V<D> res(max(size(), r.size()));
d2         REP(i, res.size()) res[i] = freq(i) - r.freq(i);
e9         return res;
67     }
ea     Poly operator*(P r) const { return {multiply(v, r.v)}; }
23     Poly operator*(D r) const {
6e         V<D> res = v;
3c         REP(i, size()) res[i] *= r;
36         return res;
55     }
77     Poly operator/(D r) const { return *this * r.inv(); }
79     Poly operator/(Poly r) const {
19         if (size() < r.size()) return {};
c0         r.shrink();
52         int n = size() - r.size() + 1;
8a         return (rev().pre(n) * r.rev().inv(n)).rev(n); // 変更
b3     }
66     Poly operator%(P r) const {
57         auto x = *this - *this / r * r;
ce         x.shrink();
f4         return x;
dc     }
45     Poly& operator+=(P r) { return *this = *this + r; }
92     Poly& operator-=(P r) { return *this = *this - r; }
d3     Poly& operator*=(P r) { return *this = *this * r; }
```

```
15     Poly& operator*=(D r) { return *this = *this * r; }
02     Poly& operator/=(P r) { return *this = *this / r; }
f3     Poly& operator/=(D r) { return *this = *this / r; }
73     Poly& operator%=(P r) { return *this = *this % r; }
a7     Poly clip(int l = 0, int r = -(1 << 30)) const {
64         if (r == -(1 << 30)) r = size();
5a         if (r < l) r = l;
14         V<D> res(r - l);
08         REP(i, r - l) res[i] = freq(i + l);
f1         return res;
51     }
81     Poly pre(int le) const { return clip(0, le); }
25     Poly rev(int n = -1) const {
54         auto res = v;
9c         if (n != -1) res.resize(n);
4f         reverse(res.begin(), res.end());
d4         return res;
ae     }
cb     Poly diff() const {
29         auto res = clip(1).v;
a8         REP(i, res.size()) res[i] *= i + 1;
d0         return res;
64     }
68     Poly inte() const {
ba         auto res = clip(-1).v;
0c         REP(i, res.size()) if (i) res[i] /= i;
23         return res;
1f     }
// f * f.inv() = 1 + g(x)x^m
65     Poly inv(int m) const {
e1         Poly f = Poly({D(1) / freq(0)});
b0         for (int i = 1; i < m; i *= 2)
3b             f = (f * D(2) - f * f * pre(2 * i)).pre(2 * i);
12         return f.pre(m);
ea     }
d6     Poly exp(int n) const {
91         assert(freq(0) == 0);
7b         Poly f({1}), g({1}), q;
84         for (int i = 2; i < n * 2; i *= 2) {
9e             g = (g * 2 - f * g * g).pre(i / 2);
46             q = pre(i / 2).diff();
a7             q += (g * (f.diff() - f * q)).pre(i - 1);
03             f += (f * (pre(i) - q.inte())).pre(i);
6a         }
67         return f.pre(n);
15     }
ad     Poly log(int n) const {
29         assert(freq(0) == 1);
39         auto f = pre(n);
58         return (f.diff() * f.inv(n - 1)).pre(n - 1).inte();
b1     }
36     friend ostream& operator<<(ostream& os, P p) {
e9         if (p.size() == 0) return os << "0";
cc         for (auto i = 0; i < p.size(); i++) {
c9             if (p.v[i]) {
b3                 os << p.v[i] << "x^" << i;
36                 if (i != p.size() - 1) os << "+";
fb             }
fb         }
b1         return os;
38     }
d8     };
```

fps_composition.hpp

md5: 9c217c

```
// f(g(x)) mod x^N (g[0] = 0)
71 template <class D>
f1 V<D> fps_composition(V<D> f, V<D> g, ll N) {
b8     ll z = 1;
06     while (z < N) z *= 2;
4e     V<D> p(z * 2), q(z * 3), res(N);
6d     REP(i, f.size()) p[i * 2] = f[i];
b8     REP(i, min<ll>(g.size(), z)) q[z * 2 + i] = -g[i];
dd     q[0] = 1;
84     auto dfs = [&](auto& h, V<D> q, ll u, ll v) -> void {
e8         if (v == z) return;
2c         ll nu = (u + 1) / 2, nv = v * 2;
79         V<D> nq = q, y(nu * 2 * (nv + 1));
81         REP(i, q.size()) if (i % 2) nq[i] = -nq[i];
62         q = multiply(q, nq);
1c         REP(d, nv + 1)
4d             REP(i, nu) y[d * nu * 2 + i] = q[d * u * 2 + i * 2];
89         h(h, move(y), nu, nv);
```

```
92     V<D> x(u * 2 * (v * 2 + 1) - 1);
29     REP(d, nv)
bd     REP(i, nu) x[d * u * 2 + i * 2 + 1] =
45         p[d * nu * 2 + i];
7a     p = tr_multiply(nq, x);
e4 };
8f     dfs(dfs, move(q), z, 1);
8c     REP(i, N) res[i] = p[z - 1 - i];
da     return res;
9c }
```

fps_power_projection.hpp

md5: c000a5

```
// [x^K] f(x) / (1 - y g(x)) mod y^N (g[0] = 0)
71 template <class D>
0a V<D> power_projection(V<D> f, V<D> g, ll K, ll N) {
14     ll u = K + 1, v = 1;
12     V<D> p(u * 2 * v), q(u * 2 * (v + 1)), res(N);
8e     REP(i, min<ll>(f.size(), u)) p[i] = f[i];
f4     REP(i, min<ll>(g.size(), u)) q[u * 2 + i] = -g[i];
4f     q[0] = 1;
7a     while (K || v < N) {
03         ll nu = (u + 1) / 2, nv = v * 2;
e2         V<D> nq = q, x(nu * 2 * nv), y(nu * 2 * (nv + 1));
89         REP(d, v + 1)
29             REP(i, u) if (i % 2) nq[d * u * 2 + i] *= -1;
e7         q = multiply(q, nq);
21         p = multiply(p, nq);
19         REP(d, nv)
db         REP(i, nu) x[d * nu * 2 + i] =
6d             p[d * u * 2 + i * 2 + K % 2];
88         REP(d, nv + 1)
cb         REP(i, nu) y[d * nu * 2 + i] = q[d * u * 2 + i * 2];
7a         u = nu, v = nv, K /= 2;
2a         swap(p, x);
3e         swap(q, y);
8b     }
c8     REP(i, N) res[i] = p[i * 2];
ac     return res;
c0 }
```

garner_convolution.hpp

md5: eaba62

```
// number-theory/modint.hpp , 同/crt.hpp (inv_mod まで) ,
// 同/garner.hpp , fps/nft.hpp
cb template <int MOD, int g>
78 V<int> multiply_mod(const V<int>& a, const V<int>& b) {
d4     V<ModInt<MOD, g>> c, d;
f8     for (auto x : a) c.push_back(x);
7b     for (auto x : b) d.push_back(x);
bd     c = multiply(c, d);
7f     V<int> res;
a9     for (auto x : c) res.push_back(x.v);
1b     return res;
fa }
b7 template <class mint>
ef V<mint> multiply_garner(V<mint> a, V<mint> b) {
1a     constexpr int m[] = {167772161, 469762049, 754974721};
00     auto g = garner<mint>(V<ll>(m, m + 3));
56     V<int> c(a.size()), d(b.size());
89     REP(i, a.size()) c[i] = a[i].v;
bf     REP(i, b.size()) d[i] = b[i].v;
41     V<V<int>>> res(3);
da     res[0] = multiply_mod<m[0], 3>(c, d);
de     res[1] = multiply_mod<m[1], 3>(c, d);
c3     res[2] = multiply_mod<m[2], 11>(c, d);
a6     V<ll> r(3);
72     V<mint> ans(res[0].size());
23     REP(i, res[0].size()) {
70         REP(j, 3) r[j] = res[j][i];
e3         ans[i] = g(r);
08     }
e2     return ans;
ea }
```

nft.hpp

md5: 4d52c9

```
// 167772161      3      5 * 2^25 + 1
// 377487361      7      45 * 2^23 + 1
// 469762049      3      7 * 2^26 + 1
// 595591169      3      71 * 2^23 + 1
// 645922817      3      77 * 2^23 + 1
```

```
// 754974721      11      45 * 2^24 + 1
// 880803841      26      105 * 2^23 + 1
// 897581057      3      107 * 2^23 + 1
// 998244353      3      119 * 2^23 + 1
5d template <class Mint>
87 void nft(bool type, V<Mint>& a) {
f2     int n = int(a.size()), s = 0;
31     while ((1 << s) < n) s++;
19     assert(1 << s == n);
cd     static V<Mint> ep;
b5     while (int(ep.size()) <= s) {
dd         ep.push_back(Mint::G.pow(Mint(-1).v >> ep.size()));
b5     }
cc     V<Mint> b(n);
d1     REP(i, s) {
12         int w = 1 << (s - i - 1);
c6         Mint now = 1;
22         for (int y = 0; y < n / 2; y += w) {
7a             REP(x, w) {
7d                 auto l = a[y << 1 | x];
7a                 auto r = now * a[y << 1 | x | w];
56                 b[y | x] = l + r;
e0                 b[y | x | n >> 1] = l - r;
1d             }
01             now *= ep[i + 1];
c9         }
13         swap(a, b);
b4     }
cc     if (type) reverse(a.begin() + 1, a.end());
1a }
80 template <class Mint>
ac V<Mint> cyconv(V<Mint> a, V<Mint> b, int k) {
75     int z = 1;
60     while (z < k) z *= 2;
c1     a.resize(z);
ec     b.resize(z);
8b     nft(0, a);
17     nft(0, b);
b7     Mint iz = Mint(z).inv();
4c     REP(i, z) a[i] *= b[i] * iz;
f7     nft(1, a);
14     a.resize(k);
39     return a;
2e }
1e template <class Mint>
1c V<Mint> multiply(V<Mint> a, V<Mint> b) {
df     int n = a.size(), m = b.size();
70     if (!n || !m) return {};
ad     if (min(n, m) <= 32) {
b7         V<Mint> ans(n + m - 1);
df         REP(i, n) REP(j, m) ans[i + j] += a[i] * b[j];
63         return ans;
03     }
2f     return cyconv(move(a), move(b), n + m - 1);
44 }
// multieval, boston_mori, composition
4e template <class Mint>
02 V<Mint> tr_multiply(V<Mint> a, V<Mint> b) {
ae     int n = a.size(), m = b.size();
e0     reverse(b.begin(), b.end());
19     b = cyconv(move(a), move(b), m);
f0     reverse(b.begin(), b.end());
17     b.resize(m - n + 1);
83     return b;
4d }
```

relaxed_convolution.hpp

md5: ed3c47

```
// f(i,c)はc=C[i-1]の時にA[i],B[i]を返す関数 (i=0の時はcは何でもよい)
71 template <class D>
06 V<D> relaxed_convolution(int N, auto f) {
3b     V<D> R(N, 0), A(N), B(N), C;
ec     D c = 0;
af     REP(i, N) {
c0         tie(A[i], B[i]) = f(i, c);
d0         for (int p = 1; (i + 2) % p == 0; p *= 2) {
2a             if (p == i + 2) break;
bb             auto calc = [&](int la, int ra, int lb, int rb) {
58                 C = multiply(V<D>(A.begin() + la, A.begin() + ra),
a1                     V<D>(B.begin() + lb, B.begin() + rb));
2b                 REP(j, C.size())
5e                     if (la + lb + j < N) R[la + lb + j] += C[j];
```

```
23     };
24     if (p * 2 != i + 2)
56         calc(p - 1, 2 * p - 1, i + 1 - p, i + 1);
c8         calc(i + 1 - p, i + 1, p - 1, 2 * p - 1);
e9     }
79     c = R[i];
3c     }
7f     return R;
ed }
```

taylor_shift.hpp

md5: 6d3b60

```
// f(x+c)を返す
5d template <class Mint>
97 V<Mint> taylor_shift(V<Mint> a, Mint c) {
40     int n = a.size();
ae     V<Mint> b(n + 1, 1), fi(n);
e2     Mint f = 1;
76     REP(i, n) {
95         a[i] *= f;
30         f *= i + 1;
98         b[i + 1] = b[i] * c;
4d     }
77     f = f.inv();
04     REP(i, n) b[n - 1 - i] *= fi[n - 1 - i] = f *= n - i;
44     reverse(b.begin(), b.end());
be     a = multiply(a, b);
8f     REP(i, n) fi[i] *= a[i + n];
dd     return fi;
6d }
```

convex

dynamic_convex_hull.hpp

md5: 95e223

```
72 struct Line { // f(x) = k * x + m
14     mutable ll k, m, p;
0c     bool operator<(const Line& o) const { return k < o.k; }
0d     bool operator<(ll x) const { return p < x; }
7e };
74 struct LineContainer : multiset<Line, less<>> {
    // (for doubles, use inf = 1/.0, div(a,b) = a/b)
4a     const ll inf = LLONG_MAX;
5b     ll div(ll a, ll b) { // floored division
3b         return a / b - ((a ^ b) < 0 && a % b);
ae     }
65     bool isect(iterator x, iterator y) {
ef         if (y == end()) {
bc             x->p = inf;
b1             return false;
1f         }
2f         if (x->k == y->k)
34             x->p = x->m > y->m ? inf : -inf;
2f         else
2c             x->p = div(y->m - x->m, x->k - y->k);
b5         return x->p >= y->p;
53     }
de     void add(ll k, ll m) {
f7         auto z = insert({k, m, 0}), y = z++, x = y;
81         while (isect(y, z)) z = erase(z);
60         if (x != begin() && isect(--x, y))
37             isect(x, y = erase(y));
a6         while ((y = x) != begin() && (--x)->p >= y->p)
ce             isect(x, erase(y));
da     }
0f     ll query(ll x) { // max f(x)
ba         assert(!empty());
e5         auto l = *lower_bound(x);
63         return l.k * x + l.m;
2f     }
95 };
```

monge_d_edge_shortest.hpp

md5: 89b418

```
// Int f(int a, int b){ return A[a][b] } (A : monge)
// void det(int i, int j, Int x){
//     argmin[i] = j; mimval[i] = x; }
2d template <class Int, class F1, class F2>
92 void easy_larsch(int n, F1 f, F2 det, Int inf) {
3c     V<int> a(n, 0);
22     V<Int> b(n, inf);
15     n--;
```

```
eb     auto check = [&](int i, int k, int d) {
72         Int v = f(i, k);
ae         if (v < b[i]) a[i] = k, b[i] = v;
ef     };
1b     auto dfs = [&](auto& dfs, int l, int r, int d) -> void {
2e         if (r - l == 1) return (void)det(r, a[r], b[r]);
45         int m = (l + r) / 2;
fc         for (int k = a[l]; k <= a[r]; k++) check(m, k, d);
8e         dfs(dfs, l, m, d + 1);
b5         for (int k = l + 1; k <= m; k++) check(r, k, d);
bd         dfs(dfs, m, r, d + 1);
32     };
4f     check(n, 0, 0);
9c     dfs(dfs, 0, n, 1);
9a }
// Int f(int a, int b){ return dist(a,b) } (dist : monge)
62 template <class Int, class F1>
db Int d_edge_shortest_path(int n, ll k, F1 f, Int max_weight,
44     Int INF) {
da     Int inf = max_weight; // 辺の重みより大きい値
70     auto DC = [&](Int p) -> pair<Int, ll> {
e4         vector<Int> D(n, INF);
97         vector<int> P(n);
f1         D[0] = 0;
20         easy_larsch(
20             n,
26             [&](int a, int b) { return D[b] + f(b, a) - p; },
2f             [&](int a, int b, Int c) { P[a] = b, D[a] = c; },
74             INF);
a6         ll c = 0, v = n - 1;
86         for (; v; v = P[v]) c++;
a2         return {D[n - 1], c};
13     };
ef     Int m = -inf + 1; // 辺の重みが正なら0にしてもいい
5f     Int ok = m, ng = inf;
69     while (abs(ok - ng) > 1) {
1d         auto mid = (ok + ng) / 2;
6b         auto [d, c] = DC(mid);
86         (c <= k ? ok : ng) = mid;
b8     }
a7     if (ok == m) return -INF;
36     auto [d, c] = DC(ok);
f6     return d + ok * k;
89 }
```

cht_monotone.hpp

md5: 91223f

```
36 template <class Pos, class F>
ce struct CHTMonotone {
45     struct Q {
2e         Pos l, r;
31         F f;
1b     };
46     deque<Q> a;
b7     CHTMonotone(Pos l, Pos r, F inf) : a({{l, r, inf}}) {}
df     void add(F f, auto fx, auto cross) {
9b         Pos l = a[0].l, r = a.back().r;
67         Q qf = {l, r, f};
f5         if (fx(f, r) < fx(a.back().f, r)) {
c6             do {
ea                 Q& q = a.back();
d7                 if (fx(f, q.l) < fx(q.f, q.l)) continue;
d5                 cross(q.f, f, q.r = q.l, qf.l = r);
9d                 break;
e7             } while (a.pop_back(), a.size());
af             a.push_back(qf);
6e         } else if (fx(f, l) < fx(a[0].f, l)) {
93             do {
27                 Q& q = a.front();
3e                 if (fx(f, q.r) < fx(q.f, q.r)) continue;
26                 cross(f, q.f, qf.r = l, q.l = q.r);
10                 break;
bc             } while (a.pop_front(), a.size());
7b             a.push_front(qf);
e5         }
11     }
33     F query(Pos x) {
84         return ranges::partition_point(a,
e3             [x](Q& f){ return f.r < x; })->f;
56     }
50     F query_inc(Pos x) {
f0         while (a[0].r < x) a.pop_front();
3d         a[0].l = x;
```

```
6e     return a[0].f;
40 }
36 F query_dec(Pos x) {
25     while (x < a.back().l) a.pop_back();
10     a.back().r = x;
ab     return a.back().f;
e6 }
91 };
// struct F { ll a, b; };
// auto fx = [&](F f, ll x){ return f.a * x + f.b; };
// void cross(F a, F b, ll& xa, ll& xb){
//     while(abs(xa - xb) > 1){
//         Pos x = xa + (xb - xa) / 2;
//         (a(x) < b(x) ? xa : xb) = x;
//     }
// }
// void cross(F a, F b, double& xa, double& xb){
//     xa = xb = -(a.b - b.b) / (a.a - b.a);
// }
```

graph

dominator_tree.hpp

md5: a10eaa

```
b7 struct DominatorTree {
60     int n, cs;
1d     V<V<int>>> E, RE, rdom;
4b     V<int> S, RS, par, val, sdom, rp, idom;
fa     DominatorTree(int n)
0b         : n(n), cs(0), E(n), RE(n), rdom(n) {
5b         S = RS = par = val = sdom = rp = idom = V<int>(n, -1);
81     }
74     void add_edge(int x, int y) { E[x].push_back(y); }
79     int find(int x, int c = 0) {
88         if (par[x] == x) return c ? -1 : x;
1b         int &w = par[x], p = find(w, 1);
20         if (p < 0) return c ? w : val[x];
be         if (sdom[val[x]] > sdom[val[w]]) val[x] = val[w];
c4         w = p;
f1         return c ? p : val[x];
37     }
db     void dfs(int x) {
9f         par[cs] = sdom[cs] = val[cs] = cs;
89         RS[S[x]] = cs++;
9e         for (int e : E[x]) {
80             if (S[e] < 0) dfs(e), rp[S[e]] = S[x];
12             RE[S[e]].push_back(S[x]);
f6         }
b6     }
27     V<int> solve(int src) { // idom[v] : dom tree で v の親
07         dfs(src);
45         for (int i = cs - 1; i >= 0; i--) {
2b             for (int e : RE[i]) chmin(sdom[i], sdom[find(e)]);
06             if (i) rdom[sdom[i]].push_back(i);
99             for (int e : rdom[i]) {
62                 int p = find(e);
7a                 idom[RS[e]] = RS[sdom[p] == i ? i : p];
e7             }
dd             if (i) par[i] = rp[i];
df         }
2a         REP(i, cs)
7b         if (i && RS[sdom[i]] != idom[RS[i]])
2c             idom[RS[i]] = idom[idom[RS[i]]];
54         return idom;
d1     }
a1 };
```

assignment_problem.hpp

md5: a097c3

```
    // result[i] : use A[i][result[i]]
e2 using Cost = ll;
8b V<int> AssignmentProblem(int n, int m, const V<V<Cost>>> A,
67                             int K, Cost INF) {
63     V<Cost> d(m + 1, 0), h(m, 0);
50     V<int> l(n, -1), r(m, -1);
a7     while (K--) {
79         V<int> f(m), par(m, -1);
4e         auto ch = [&](int w, Cost f, int v) {
47             if (f < d[w]) d[w] = f, par[w] = v;
8a         };
f7         REP(v, m + 1) d[v] = INF;
fa         REP(v, n)
```

```
b7     if (l[v] < 0) REP(w, m) ch(w, A[v][w] - h[w], v);
f3     int w = -1;
71     while (1) {
4e         w = m;
8e         REP(i, m) if (!f[i] && d[i] < d[w]) w = i;
01         f[w] = 1;
34         if (r[w] < 0) break;
5c         int v = r[w];
85         auto c = d[w] + h[w] - A[v][w];
44         REP(x, m) if (!f[x]) ch(x, c + A[v][x] - h[x], v);
f0     }
0c     REP(i, m) if (f[i]) h[i] += d[i] - d[w];
0e     while (w >= 0) swap(w, l[r[w] = par[w]]);
cb }
fd     return l;
a0 }
```

biconnectivity.hpp

md5: 64967f

```
17 struct Biconnectivity {
2d     V<pair<int, int>> bct; // ( 2-連結成分の番号, 頂点 )
d3     V<int> tecc; // tecc[頂点] == 2辺連結成分の番号
a7     int num_bcc, num_tecc;
5b     Biconnectivity(int n, const V<V<int>>& g) : tecc(n) {
a9         V<int> low(n), ord(n, -1), st(n), est(n + 1, n);
92         int it = 0, idx = 0, eit = 0, f = 0;
e5         auto dfs = [&](auto& dfs, int v, int p) -> void {
96             st[it] = est[eit++] = v;
ba             low[v] = ord[v] = it++;
1e             for (int w : g[v])
3f                 if (w != p || !(p = -1)) {
2f                     if (ord[w] < 0) {
ed                         dfs(dfs, w, v);
a3                         low[v] = min(low[v], low[w]);
2a                         if (low[w] >= ord[v]) {
e4                             while (ord[w] < it)
cc                                 bct.push_back({idx, st[--it]});
56                             bct.push_back({idx++, v});
f5                         }
ef                     }
33                     low[v] = min(low[v], ord[w]);
da                 }
b4                 if (low[v] == ord[v] && ++f)
46                     while (est[eit] != v) tecc[est[--eit]] = f - 1;
31             };
3b             REP(v, n) if (ord[v] < 0) dfs(dfs, v, -1);
6f             REP(v, n) if (!g[v].size()) bct.push_back({idx++, v});
ae             num_bcc = idx;
79             num_tecc = f;
1c         }
64     };
```

cliques.hpp

md5: 3739d9

```
    // 計算量は O( V^2 + V 2 ^ sqrt(2E) )
    // 空集合は含まれない
    // void f(const V<int>& clique);
25 void cliques(int n, V<V<int>>> graph, auto f) {
07     V<V<int>>> g(n, V<int>(n));
2d     REP(x, n) for (auto y : graph[x]) g[x][y] = 1;
b9     V<int> deg(n);
a2     REP(i, n) REP(j, n) deg[i] += g[i][j];
e6     REP(it, n) {
71         int v = 0;
69         REP(i, n) if (deg[i] < deg[v]) v = i;
14         V<int> H, C = {v}, F(deg[v]);
ec         for (int w : graph[v])
ce             if (deg[w]-- < n) H.push_back(w);
66         REP(i, deg[v]) REP(j, i) F[j] |= g[H[i]][H[j]] << i;
aa         auto dfs = [&](auto& dfs, int m) -> void {
13             f(C);
05             REP(i, deg[v]) if (m & 1 << i) {
a6                 C.push_back(H[i]);
bf                 dfs(dfs, (m -= 1 << i) & F[i]);
f9                 C.pop_back();
6f             }
79         };
21         dfs(dfs, (1 << deg[v]) - 1);
83         deg[v] = n * 5;
d7     }
37 }
```

directed_mst.hpp

md5: 4bc9ce

```

cb  template <typename T, bool isMin = true>
bd  struct LeftistHeap {
ec      struct Node {
9d          T key, lazy;
cf          Node *l, *r;
f1          int s;
98          int idx;
e3          Node(T key, int idx)
7e              : key(key), lazy(0), l(0), r(0), s(1), idx(idx) {}
61      };
db      using np = Node*;
36      np alloc(T key, int idx) { return new Node(key, idx); }
2f      np add(np t, T z) {
26          if (t) t->lazy += z, t->key += z;
c8          return t;
39      }
a9      np prop(np t) {
99          t->l = add(t->l, t->lazy);
8d          t->r = add(t->r, t->lazy);
a5          t->lazy = 0;
6d          return t;
be      }
43      np meld(np a, np b) {
32          if (!a || !b) return a ? a : b;
ca          a = prop(a), b = prop(b);
32          if ((a->key < b->key) ^ isMin) swap(a, b);
bb          a->r = meld(a->r, b);
79          if (!a->l || a->l->s < a->r->s) swap(a->l, a->r);
ea          a->s = (a->r ? a->r->s : 0) + 1;
82          return a;
4d      }
49      np pop(np t) { return meld(t->l, t->r); }
bf  };

```

```

b8  struct Edge {
40      using Cost = ll;
2b      int from, to;
27      Cost cost;
f1  };
// root -----> leaf
// from -----> to
// ans[root] = -1; ans[v] = (? --> v) なる辺番号
73  V<int> directed_mst(int n, int root, V<Edge> E) {
ac      REP(i, n) if (i != root) E.push_back({(int)i, root, 0});
86      int x = 0;
3a      V<int> par(2 * n, -1), vis(par), uf(par), st;
3f      using Heap = LeftistHeap<Edge::Cost, 1>;
c0      Heap h;
70      V<typename Heap::np> ins(2 * n, 0);
07      REP(i, E.size())
23          ins[E[i].to] =
69              h.meld(ins[E[i].to], h.alloc(E[i].cost, i));
d2      auto Find = [&](auto& f, int a) -> int {
ea          return uf[a] < 0 ? a : (uf[a] = f(f, uf[a]));
24      };
e4      auto go = [&](int v) {
3b          return Find(Find, E[ins[v]->idx].from);
45      };
49      for (int i = n; ins[x]; i++) {
56          for (; vis[x] == -1; x = go(x)) vis[x] = 0;
cb          for (; x != i; x = go(x)) {
d6              auto w = ins[x]->key;
4d              ins[i] = h.meld(ins[i], h.add(h.pop(ins[x]), -w));
13              uf[x] = par[x] = i;
62          }
49          for (; ins[x] && go(x) == x; ins[x] = h.pop(ins[x]))
71              ;
4e      }
f5      V<int> ans(n, -1);
aa      for (int i = root; i != -1; i = par[i]) vis[i] = 1;
64      do {
ba          if (vis[x] != 1) {
a0              int e = ins[x]->idx;
63              ans[E[e].to] = e;
aa              for (int j = E[e].to; j != -1 && !vis[j]; j = par[j])
63                  vis[j] = 1;
41          }
db      } while (x--);
65      return ans;
4b  }

```

eulerian_trail.hpp

md5: 625dd3

```

// オイラー路を一つ構築して { 頂点列 , 辺列 } を返す
43  pair<V<int>, V<int>> eulerian_trail(
4e      int N, const V<pair<int, int>>& edges, bool directed,
9d      int start = -1) {
b1      int M = edges.size();
14      if (!M) return {{0}, {}};
e9      V<int> X(M), D(N);
b7      V<V<int>> G(N);
b8      REP(m, M) {
37          auto [u, v] = edges[m];
21          G[u].push_back(m);
c5          if (!directed) G[v].push_back(m);
19          X[m] = u + v;
69      }
53      if (start < 0) {
bb          start = edges[0].first;
01          for (auto [u, v] : edges) {
d3              D[u] += 1;
ec              D[v] += directed ? -1 : 1;
95          }
ad          REP(i, N) if (D[i] % 2 == 1) start = i;
ba      }
88      V<int> H(M + 1, start), E(M), U(M, 0), K(N);
73      REP(i, N) K[i] = G[i].size();
0e      int p = 0, q = M;
b6      while (q) {
fa          int v = H[p];
7e          while (0 < K[v]-- && U[G[v]][K[v]])
9d              ;
05          if (K[v] < 0) {
ed              if (!p) return {{-1}, {-1}};
6d              H[q--] = H[p--];
97              E[q] = E[p];
10          } else {
09              auto e = G[v][K[v]];
86              U[e] = 1;
21              E[p++] = e;
e6              H[p] = X[e] - v;
e5          }
13      }
2f      REP(i, M)
c0      if (X[E[i]] != H[i] + H[i + 1]) return {{-1}, {-1}};
d8      return {H, E};
62  }

```

manhattan_mst.hpp

md5: ea5b58

```

// 候補の辺を O(N) 本に減らす。MST時は追加でsort,
// UF等の必要あり。
67  template <typename T>
9a  V<tuple<T, int, int>> manhattan_mst(V<T> xs, V<T> ys) {
52      assert(xs.size() == ys.size());
c1      V<tuple<T, int, int>> ret;
5d      int n = (int)xs.size();
f6      V<int> ord(n);
be      REP(i, n) ord[i] = i;
65      auto cmp = [&](int i, int j) {
b9          return xs[i] + ys[i] < xs[j] + ys[j];
6d      };
4d      REP(s, 4) {
d3          sort(ord.begin(), ord.end(), cmp);
10          map<T, int> idx;
8f          for (int i : ord) {
d1              for (auto it = idx.lower_bound(-ys[i]);
b6                  it != idx.end(); it = idx.erase(it)) {
06                  int j = it->second;
3e                  if (xs[i] - xs[j] < ys[i] - ys[j]) break;
63                  ret.emplace_back(
0c                      abs(xs[i] - xs[j]) + abs(ys[i] - ys[j]), i, j);
50              }
b2              idx[-ys[i]] = i;
60          }
d6          swap(xs, ys);
b7          if (s % 2) REP(i, n) xs[i] *= -1;
b7      }
99      return ret;
ea  }

```

max_flow.hpp

md5: 0e6583

```
/*
struct E {
    int to, rev, cap;
};
V<V<E>> g;
auto add_edge = [&](int from, int to, int cap) {
    g[from].push_back(E{to, int(g[to].size()), cap});
    g[to].push_back(E{from, int(g[from].size())-1, 0});
};
*/
40 template <class C>
ab struct MaxFlow {
2a     C flow;
d3     V<char> dual; // false: S-side true: T-side
b3 };
6d template <class C, class E>
8e struct MFExec {
cd     static constexpr C INF = numeric_limits<C>::max();
91     C eps;
a8     V<V<E>>& g;
e0     int s, t;
c5     V<int> level, iter;
26     C dfs(int v, C f) {
ae         if (v == t) return f;
15         C res = 0;
64         for (int& i = iter[v]; i < int(g[v].size()); i++) {
d7             E& e = g[v][i];
60             if (e.cap <= eps || level[v] >= level[e.to])
14                 continue;
3d             C d = dfs(e.to, min(f, e.cap));
b2             e.cap -= d;
36             g[e.to][e.rev].cap += d;
63             res += d;
26             f -= d;
04             if (f == 0) break;
24         }
ff         return res;
50     }
26     MaxFlow<C> info;
MFExec(V<V<E>>& _g, int _s, int _t, C _eps)
59         : eps(_eps), g(_g), s(_s), t(_t) {
ee         int N = int(g.size());
a1         C& flow = (info.flow = 0);
b6         while (true) {
d5             queue<int> que;
37             level = V<int>(N, -1);
db             level[s] = 0;
75             que.push(s);
39             while (!que.empty()) {
fc                 int v = que.front();
bc                 que.pop();
c5                 for (E e : g[v]) {
91                     if (e.cap <= eps || level[e.to] >= 0) continue;
d1                     level[e.to] = level[v] + 1;
3d                     que.push(e.to);
d4                 }
2c             }
f2             if (level[t] == -1) break;
2d             while (true) {
d2                 iter = V<int>(N, 0);
86                 C f = dfs(s, INF);
6c                 if (!f) break;
0c                 flow += f;
5f             }
d4         }
e0         for (int i = 0; i < N; i++)
1e             info.dual.push_back(level[i] == -1);
4f     }
f4 };
4e template <class C, class E>
7a MaxFlow<C> get_mf(V<V<E>>& g, int s, int t, C eps) {
7b     return MFExec<C, E>(g, s, t, eps).info;
0e }
```

max_matching-daimonge.hpp

md5: aa4a41

```
aa V<pair<ll, ll>> MaxMatching(ll n, V<pair<ll, ll>> &es) {
50     mt19937 rnd;
f8     V<V<ll>> g(n + 1);
bb     for (auto &[u, v] : es) {
```

```
52         g[u + 1].push_back(v + 1);
95         g[v + 1].push_back(u + 1);
c5     }
cd     V<ll> used(n + 1), mate(n + 1);
ce     ll T;
29     auto dfs = [&](auto &dfs, ll v) -> bool {
d7         used[v] = T;
ca         shuffle(g[v].begin(), g[v].end(), rnd);
15         for (auto &u : g[v]) {
8e             ll w = mate[u];
69             if (used[w] < T) {
ed                 mate[v] = u, mate[u] = v, mate[w] = 0;
06                 if (!w or dfs(dfs, w)) return 1;
c4                 mate[u] = w, mate[w] = u, mate[v] = 0;
26             }
58         }
a5         return 0;
50     };
4b     REP(_, 10) {
7e         used.assign(n + 1, 0);
b8         REP(v, n) if (!mate[v + 1]) T++, dfs(dfs, v + 1);
49     }
ca     V<pair<ll, ll>> ret;
84     REP(v, n)
94         if (v + 1 < mate[v + 1])
62             ret.push_back({v, mate[v + 1] - 1});
11     return ret;
aa }
```

min_cost_flow.hpp

md5: 3550f7

```
// 辺は配列 E に一列に並んでいる。
// 辺 e (e : int) の逆辺は、辺 e^1 である。
dc struct MCF {
1c     using Cap = ll;
3a     using Cost = ll;
a0     Cost INF = 1ll << 60;
c5     struct Edge {
cf         int from;
e2         int to;
86         Cost cost;
06         Cap cap;
fc     };
22     int N;
77     V<Edge> E;
37     V<V<int>> g;
c6     V<Cost> P;
6e     MCF(int n) : N(n), g(n), P(n, 0) {}
b9     int addEdge(int u, int v, Cap cap, Cost cost) {
01         int e = E.size();
42         g[u].push_back(e);
7b         g[v].push_back(e + 1);
c5         E.push_back({u, v, cost, cap});
d2         E.push_back({v, u, -cost, 0});
e9         return e;
fc     }
01     pair<Cap, Cost> flowuni(int s, int t, Cap lim) {
a7         V<Cost> dist(N, INF);
8a         V<int> par(N, -1);
45         V<Cap> fl(N, lim);
85         priority_queue<pair<Cost, int>> que;
e9         dist[s] = 0;
08         que.push({0, s});
bd         while (que.size()) {
8c             auto [d, v] = que.top();
6a             que.pop();
98             if (dist[v] != -d) continue;
19             if (v == t) break;
bd             for (auto e : g[v])
1b                 if (E[e].cap) {
97                     int w = E[e].to;
ac                     Cost nd = -d + P[v] + E[e].cost - P[w];
df                     if (dist[w] <= nd) continue;
83                     dist[w] = nd;
e3                     que.push({-nd, w});
28                     par[w] = e;
ee                     fl[w] = min(fl[v], E[e].cap);
8d                 }
37             }
4b             if (par[t] < 0) return {0, 0};
11             lim = fl[t];
82             for (int p = t; p != s;) {
ca                 int e = par[p];
```

```
7f      E[e].cap -= lim;
a7      E[e ^ 1].cap += lim;
dd      p = E[e ^ 1].to;
16    }
4e    for (int i = 0; i < N; i++)
14      if (i == s || par[i] >= 0) {
f9        P[i] += dist[i] - dist[t];
6e      }
00    return {P[t] - P[s], lim};
d7  }
f8  V<pair<Cap, Cost>> slope(int s, int t, Cap lim) {
92    V<pair<Cap, Cost>> res;
d4    res.push_back({0, 0});
04    Cap fl = 0;
2f    Cost co = 0;
f7    while (fl < lim) {
87      auto [u, v] = flowuni(s, t, lim - fl);
f0      if (v == 0) break;
df      res.push_back({fl += v, co += u * v});
69    }
0a    return res;
95  }
fd  pair<Cap, Cost> flow(int s, int t, Cap lim) {
b8    return slope(s, t, lim).back();
bb  }
35  };
```

scc.hppmd5: 467325

```
// res.first = 強連結成分の個数
// res.second[v] = v が属する強連結成分の番号
be pair<int, V<int>> scc(int n, V<V<int>> g) {
e1   int tm = 0, it = 0, idx = 0;
e8   V<int> ord(n, -1), st(n + 1, n), id(n, n);
16   auto dfs = [&](auto& dfs, int v) -> int {
98     int low = ord[v] = tm++;
ba     st[it++] = v;
39     for (auto w : g[v])
19       if (id[w] >= n)
6e         chmin(low, ord[w] < 0 ? dfs(dfs, w) : ord[w]);
7a     if (low == ord[v] && ++idx)
f4       while (st[it] != v) id[st[--it]] = idx;
1e     return low;
c3   };
06   REP(i, n) if (ord[i] == -1) dfs(dfs, i);
98   for (auto& x : id) x = idx - x;
09   return make_pair(idx, move(id));
46 }
```

two-sat.hppmd5: 9301c7

```
d9 struct TwoSat {
e5   int n;
3a   V<bool> res;
f3   V<V<int>> g;
81   TwoSat(int n) : n(n), g(n * 2) {}
cb   void add(int x) { g.resize((n += x) * 2); }
47   void either(ll i, bool fi, ll j, bool fj) {
87     i = i * 2 + fi, j = j * 2 + fj;
96     g[i ^ 1].push_back(j);
40     g[j ^ 1].push_back(i);
f6   }
fe   bool solve() {
e7     auto id = scc(g.size(), g).second;
9e     res.resize(n);
16     REP(i, n) if (id[i * 2] == id[i * 2 + 1]) return 0;
35     REP(i, n) res[i] = id[i * 2] < id[i * 2 + 1];
b2     return 1;
f6   }
09   void atMostOne(V<pair<ll, bool>> a) {
b4     ll x = n, i = 0;
18     add(a.size());
1c     for (auto [v, f] : a) {
af       either(v, !f, x + i, 1);
42       if (i) either(x + i - 1, 0, x + i, 1);
14       if (i) either(v, !f, x + i - 1, 0);
58       i++;
11     }
8b   }
93 };
```

treehld.hppmd5: a85343

```
40 struct HLD {
5b   int N;
44   int g;
e6   V<V<int>> t;
88   V<int> par, dep, sz, pp, stov, vtos, px, index;
4c   HLD(int N, V<V<int>> graph, int root)
7b     : N(N), g(0), t(move(graph)), par(N, -1), dep(N),
21       sz(N, 1), pp(N, root), stov(N), vtos(N), px(N),
d5       index(N) {
ae     dfs1(root);
f6     dfs2(root);
40   }
7c   void dfs1(int v) {
b0     for (auto& w : t[v]) {
3b       t[w].erase(find(t[w].begin(), t[w].end(), v));
d1       par[w] = v;
62       dep[w] = dep[v] + 1;
86       dfs1(w);
75       sz[v] += sz[w];
03       if (sz[t[v][0]] < sz[w]) swap(t[v][0], w);
bc     }
e9   }
fc   void dfs2(int v) {
35     vtos[v] = g;
81     stov[g++] = v;
a5     for (auto w : t[v]) {
e7       int f = w == t[v][0];
a9       pp[w] = f ? pp[v] : w;
3c       px[w] = px[v] + 1 - f;
7b       dfs2(w);
10     }
0c   }
27   int lca(int a, int b) {
0c     if (px[pp[a]] > px[pp[b]]) swap(a, b);
3e     while (px[pp[a]] < px[pp[b]]) b = par[pp[b]];
c9     while (pp[a] != pp[b]) {
4c       a = par[pp[a]];
54       b = par[pp[b]];
f9     }
55     return dep[a] < dep[b] ? a : b;
f9   }
1e   auto query_half(int r, int a, bool edge = false) {
62     vector<pair<int, int>> res(px[a] - px[r] + 1);
55     while (px[a] > px[r]) {
77       res[px[a] - px[r]] = {vtos[pp[a]], vtos[a] + 1};
0c       a = par[pp[a]];
72     }
fe     res[0] = {vtos[r] + edge, vtos[a] + 1};
dd     return res;
05   }
// return vector<pair<int,int>>
// a -> b
// l<rなら[l:r)、l>rなら[r:l)の逆順
// edge=trueの場合はlcaを含まない区間を返す
dd   auto query(int a, int b, bool edge = false) {
9a     int g = lca(a, b);
c3     auto l = query_half(g, b, edge);
26     auto r = query_half(g, a, 1);
f1     for (auto& q : r) swap(q.first, q.second);
88     l.insert(l.begin(), r.rbegin(), r.rend());
2a     return l;
95   }
// 返り値は縮約後の縮約後の頂点集合 vector<int> と
// 木の辺の集合 vector<pair<int,int>> !! もとの頂点番号
// !! もし reindex したいなら index[v] を見ればいい
2c   pair<V<int>, V<pair<int, int>>> ComputeTree(V<int> vs) {
8a     auto comp = [&](int x, int y) {
8d       return vtos[x] < vtos[y];
2a     };
40     sort(vs.begin(), vs.end(), comp);
c7     int K = vs.size();
96     REP(i, K - 1) vs.push_back(lca(vs[i], vs[i + 1]));
5a     sort(vs.begin(), vs.end(), comp);
c2     vs.erase(unique(vs.begin(), vs.end()), vs.end());
88     K = vs.size();
5f     REP(i, K) index[vs[i]] = i;
46     V<pair<int, int>> es;
f8     REP(i, K - 1)
```

```

2b     es.push_back({lca(vs[i], vs[i + 1]), vs[i + 1]});
0e     return {vs, es};
bc     }
a8 };

```

link_cut_tree.hpp

md5: 20f709

```

75 struct S {
63     ll x;
41 };
94 static S op(S l, S r) { return {l.x + r.x}; }
10 struct LNode {
43     LNode *c[2] = {0, 0}, *p = 0;
f4     S a, sum;
78     int rev = 0;
09     void do_rev() { rev ^= 1, swap(c[0], c[1]), agg(); }
98     void prop() {
a7         if (rev) REP(t, 2) if (c[t]) c[t]->do_rev();
85         rev = 0;
32     }
65     void agg() {
90         sum = a;
96         if (c[0]) sum = op(c[0]->sum, sum);
2f         if (c[1]) sum = op(sum, c[1]->sum);
69     }
42     int way() {
f1         REP(t, 2) if (p && p->c[t] == this) return t;
59         return -1;
0f     }
81     void rot() {
7e         auto w = p;
4a         int x = way();
ea         if (w->way() != -1) w->p->c[w->way()] = this;
ab         p = w->p;
d6         w->c[x] = c[!x];
a8         if (c[x ^ 1]) c[!x]->p = w;
49         c[!x] = w;
a9         w->p = this;
23         w->agg(), agg();
16     }
f2     void propall() {
37         if (way() >= 0) p->propall();
22         prop();
ea     }
16     void splay() {
ab         for (propall(); way() >= 0; rot())
ef             if (p->way() != -1)
4b                 (way() == p->way() ? p : this)->rot();
0c     }
58     void expose() {
f1         for (auto t = this; t; t = t->p) t->splay();
46         for (auto t = this; t->p; t = t->p) {
ec             t->p->c[1] = t;
50             t->p->agg();
86         }
78         splay(), c[1] = 0, agg();
6d     }
94     void evert() { expose(), do_rev(); }
3a     void link(LNode* r) { r->evert(), evert(), p = r; }
e1     void cut() { expose(), c[0] = c[0]->p = 0; }
20 };

```

static_top_tree.hpp

md5: 4a3228

```

3a struct StaticTopTree {
e5     static const int VERT = 0; // -1 -1 vtx
d3     static const int COMP = 1; // root child eidxn
10     static const int RAKE = 2; // child child -1
3c     static const int ADDE = 3; // child -1 eidxn
1f     static const int ADDV = 4; // child -1 vtx
9b     using pr = pair<int, int>;
4e     int n, rt, k, num_node;
37     V<V<pr>> g;
3f     V<int> PE, vh;
fa     struct Node {
18         int p, l, r, e, t;
f5     };
55     V<Node> A;
fd     StaticTopTree() {}
d3     StaticTopTree(int n, const V<pr>& tree, int root)
e3         : n(n), rt(root), k(0), g(n), PE(n, -1),
f8         vh(n * 2, 1), A(4 * n + 30, {-1, -1, -1, 0, -1}) {
a4         REP(i, n - 1) {

```

```

0b         auto [u, v] = tree[i];
fe         g[u].push_back({v, i});
4d         g[v].push_back({u, i});
3e     }
3b     dfs(rt);
81     A.resize(num_node = comp(rt).first + 1);
25 }
9c     int dfs(int c) {
2c         int s = 1, best = 0, t = 0;
a4         for (auto& e : g[c]) {
a5             auto [w, ei] = e;
03             g[w].erase(ranges::find(g[w], pr{c, PE[w] = ei}));
b2             s += t = dfs(w);
0e             if (best < t) best = t, swap(e, g[c][0]);
2a         }
13         return s;
a0     }
99     int add(int l, int r, int e, int t) {
45         A[k] = {-1, l, r, e, t};
4c         if (l >= 0) A[l].p = k;
1e         if (r >= 0) A[r].p = k;
95         if (e >= 0) vh[e] = k;
cb         return k++;
cb     }
da     pr merge(const V<pr>& a, int t) {
91         if (a.size() == 1) return a[0];
c8         int u = 0;
2c         for (auto& [_, s] : a) u += s;
22         V<pr> b, c;
68         for (auto& [i, s] : a) {
b2             (u > s ? b : c).emplace_back(i, s);
5e             u -= s * 2;
74         }
9e         int e = t == 1 ? n + PE[A[c[0].first].e] : -1;
b8         return {
96             add(merge(b, t).first, merge(c, t).first, e, t),
0c             -u};
95     }
30     pr comp(int v) {
b4         V<pr> chs{addv(v)};
e9         while (g[v].size())
6a             chs.push_back(addv(v = g[v][0].first));
80         return merge(chs, 1);
cb     }
a2     pr rake(int v) {
69         if (g[v].size() < 2) return {-1, 0};
1b         V<pr> chs;
6a         REP(i, g[v].size() - 1)
fa             chs.push_back(adde(g[v][i + 1].first));
ab         return merge(chs, 2);
d7     }
a4     pr adde(int v) {
9a         auto [j, sj] = comp(v);
63         return {add(j, -1, n + PE[v], 3), sj};
7e     }
93     pr addv(int v) {
38         auto [i, sj] = rake(v);
1c         return {add(i, -1, v, i < 0 ? 0 : 4), ++sj};
a0     }
33     auto& operator[](int i) const { return A[i]; }
92 };
62 template <class Structure>
3e struct STTAagg {
cf     Structure d;
23     V<class Structure::Node> A;
50     StaticTopTree st;
76     void agg(int v) {
ed         auto [p, l, r, e, t] = st[v];
40         if (t == 0) A[v].s = d.vert(e);
4f         if (t == 1) A[v].s = d.comp(A[l].s, e - st.n, A[r].s);
e4         if (t == 2) A[v].p = d.rake(A[l].p, A[r].p);
f1         if (t == 3) A[v].p = d.adde(e - st.n, A[l].s);
3d         if (t == 4) A[v].s = d.addv(e, A[l].p);
ab         if (t == 5) A[v].s = d.rev(A[l].s);
9a     }
68     void aggX(int v) {
a8         while (v >= 0) agg(v), v = st[v].p;
6f     }
f3     void aggV(int v) { aggX(st.vh[v]); }
3d     void aggE(int e) { aggX(st.vh[st.n + e]); }
4b     void init() {
6e         A.resize(st.num_node);
a1         REP(i, A.size()) agg(i);

```

```
23     }
72 };
8f struct STTStructure {
9f     struct Seg {};
d2     struct Pt {};
15     struct Node {
82         Seg s;
45         Pt p;
2e     };
a3     Seg vert(int v) { return {}; }
87     Seg comp(Seg l, int e, Seg r) { return {}; }
72     Pt rake(Pt l, Pt r) { return {}; }
6a     Seg addv(int v, Pt p) { return {}; }
33     Pt adde(int e, Seg s) { return {}; }
ed     Seg rev(Seg s) { return s; }
4a };
// V<pair<int,int>> edges(N-1);
// STTAgg<STTStructureInst> ds;
// ds.d.vtx = move(A);
// ds.d.edge = move(E);
// ds.st = StaticTopTree(N, edges, 0);
// ds.init();
```

tree_dp.hppmd5: a39ad5

```
// F : S rake(S l, S r);
// G : S extend(S a, int old_root, int new_root)
86 template <class S, class F, class G>
a7 struct TreeDP {
ed     int n, root;
52     V<int> par;
33     V<S> low, high;
aa     F f;
60     G g;
8c     TreeDP(V<V<int>> t, V<S> prim, F f, G g, int r, S e)
b1         : n(t.size()), root(r), par(n, -1), low(prim),
1b         high(prim), f(move(f)), g(move(g)) {
f9         V<int> bfs(n, r);
dc         int p = 1;
7c         for (int v : bfs)
07             for (int w : t[v]) {
5f                 ranges::rotate(t[w], ranges::find(t[w], v) + 1);
4e                 t[w].pop_back();
52                 par[bfs[p++] = w] = v;
e0             }
45             REP(i, n - 1) {
21                 int v = bfs[n - 1 - i], w = par[v];
46                 low[w] = f(g(low[v], v, w), low[w]);
d8             }
9e             for (int v : bfs) {
07                 S s = prim[v];
51                 for (int w : t[v]) {
63                     high[w] = s;
25                     s = f(s, g(low[w], w, v));
50                 }
a7                 ranges::reverse(t[v]);
6b                 s = v == r ? e : g(high[v], par[v], v);
7b                 for (int w : t[v]) {
3a                     high[w] = f(s, high[w]);
6a                     s = f(g(low[w], w, v), s);
a3                 }
c4             }
df         }
c4         S get(int v) {
c0             if (v == root) return low[v];
28             return f(low[v], g(high[v], par[v], v));
af         }
a3     };
```

string

z_algorithm.hppmd5: 3956ea

```
19 template <class Iter>
63 V<int> z_algorithm(Iter s, Iter s_end) {
75     int n = s_end - s;
0d     vector<int> r(n + 1);
8f     for (int i = 1, j = 0; i <= n; i++) {
33         int& k = r[i];
87         if (i < j + r[j]) k = min(j + r[j] - i, r[i - j]);
da         while (i + k < n && s[k] == s[i + k]) k++;
1b         if (j + r[j] < i + r[i]) j = i;
```

```
05     }
18     r[0] = n;
05     return r;
39 }
```

aho-corasick-direct-link.hppmd5: 864ee9

```
d4 struct ACNode {
61     ll len = 0, li = 0; // suffix link
eb     array<ll, 26> nx = {}; // 1 文字足したときの最長接尾辞
57 };
61 template <class Str, ll offset = 'a'>
c0 pair<V<ACNode>, V<ll>> AhoCorasick(const V<Str>& s) {
10     V<ACNode> a(1);
79     V<ll> p(s.size()), bfs(s.size());
fb     REP(i, s.size()) bfs[i] = i;
69     REP(i, bfs.size()) {
04         ll v = bfs[i];
3a         if (a[p[v]].len == s[v].size()) continue;
43         ll &q = p[v], c = s[v][a[q].len] - offset;
cc         if (a[a[q].nx[c]].len <= a[q].len) {
21             a.push_back(ACNode());
a6             a[q].nx[c] = a.size() - 1;
65             a.back().len = a[q].len + 1;
51         }
7a         q = a[q].nx[c];
ad         bfs.push_back(v);
b0     }
cc     for (auto& q : a) REP(j, q.nx.size()) if (q.len) {
2c         if (q.nx[j]) a[q.nx[j]].li = a[q.li].nx[j];
c2         if (!q.nx[j]) q.nx[j] = a[q.li].nx[j];
1f     }
5e     return {a, p};
86 };
```

aho-corasick.hppmd5: 308aec

```
d1 struct ACTrie {
8e     using NP = ACTrie*;
b2     V<int> acc;
e0     map<int, NP> next;
4c     NP fail = 0, dnx = 0;

2b private:
6b     void add(const string& s, int id, int p = 0) {
d1         if (p == int(s.size())) {
b5             acc.push_back(id);
01             return;
ff         }
3b         if (!next[s[p]]) next[s[p]] = new ACTrie();
8d         next[s[p]]->add(s, id, p + 1);
a0     }
9b     template <class OP>
2a     NP count(OP op, int p) {
4e         if (!fail) return this;
23         for (int id : acc) op(id, p);
53         if (dnx) {
50             dnx->count(op, p);
4a         } else {
54             dnx = fail->count(op, p);
bd         }
2f         return acc.size() ? this : dnx;
36     }

c7 public:
// パターンにマッチするたびにop(string ID,
// 発見位置の終端)を呼び出す
// 終端が同じで複数マッチする文字列が存在する場合、長い順に呼び出さ
れる
// s = "abaaba", pattern = {"ab", "ba"} なら
// op(0, 2), op(1, 3), op(0, 5), op(1, 6)
8c     template <class OP>
e4     void match(const string& s, OP op, int p = 0) {
a5         if (p == int(s.size())) return;
05         if (next[s[p]]) {
c7             next[s[p]]->count(op, p + 1);
d3             next[s[p]]->match(s, op, p + 1);
c3         } else {
f2             if (!fail)
59                 match(s, op, p + 1); // root
cb             else
ad                 fail->match(s, op, p); // other
96         }
```

```
54     }
b5     static NP make(V<string> v) {
90         NP tr = new ACTrie();
cb         for (int i = 0; i < int(v.size()); i++) {
44             tr->add(v[i], i);
12         }
2c         V<NP> q = {tr};
c7         tr->fail = 0;
82         for (size_t qi = 0; qi < q.size(); qi++) {
56             NP ntr = q[qi];
48             for (auto [i, to] : ntr->next) {
67                 NP f = ntr->fail;
02                 while (f && !f->next.count(i)) f = f->fail;
1d                 to->fail = f ? f->next[i] : tr;
cb                 q.push_back(to);
bc             }
c3         }
ed         return tr;
11     }
30 };
```

lcp_array.hpp

md5: c04e48

```
e8 V<ll> lcp_array(auto s, V<ll> sa) {
be     ll n = s.size(), h = 0;
91     V<ll> rsa(n + 1), lcp(n);
d7     REP(i, n) rsa[sa[i]] = i;
4b     REP(i, n) {
cb         if (h > 0) h--;
a5         if (rsa[i] == n - 1) continue;
7e         ll j = sa[rsa[i] + 1];
d2         for (; j + h < n && i + h < n; h++) {
e4             if (s[j + h] != s[i + h]) break;
2b         }
e5         lcp[rsa[i]] = h;
d2     }
ba     return lcp;
c0 }
```

manacher.hpp

md5: ffe902

```
    // res[i] = ( l + r == i なる回文 s[l..r) の長さの最大値 )
    // dummy は s 内で使用されていない値
f3 template <class S>
c9 V<int> manacher(const S& s, int dummy) {
a9     int n = s.size(), m = n * 2 + 1;
98     S a(m, dummy);
9a     REP(i, n) a[i * 2 + 1] = s[i];
e8     V<int> r(m, 1);
66     for (int i = 1, j = 0; i < m; i++) {
6e         int& k = r[i];
d2         k = i < j + r[j] ? min(j + r[j] - i, r[2 * j - i]) : 0;
7d         while (0 <= i - k && i + k < m && a[i - k] == a[i + k])
65             k++;
73         if (j + r[j] < i + r[i]) j = i;
99     }
c6     REP(i, m) r[i] -= 1;
96     return r;
ff }
```

mp.hpp

md5: ed660d

```
    // S[:i]の接頭辞と接尾辞が最大何文字一致しているか
    // S[:i]=T*cとかけるとけるTの長さの最小値はi%mp[i]=0?mp[i]:i
f3 template <class S>
6e V<int> mp(const S& s) {
23     int n = int(s.size());
d3     V<int> r(n + 1);
6f     r[0] = -1;
73     for (int i = 0, j = -1; i < n; i++) {
74         while (j >= 0 && s[i] != s[j]) j = r[j];
3c         j++;
75         r[i + 1] = j;
ba     }
42     return r;
ed }
```

palindromic_tree.hpp

md5: c35e22

```
24 struct PalNode {
38     ll len = 0, diff = 0;
d8     PalNode *li = 0;    // suffix link
```

```
e7     PalNode *par = 0;    // abcba --> bcb
fe     PalNode *sli = 0;   // series link
61     PalNode *nx[26] = {};
1d };
1a V<PalNode> pool(1000000); // string length + 0(1)
43 PalNode *poolptr = pool.data();
02 struct PalTree {
ea     using np = PalNode *;
ca     np n0, n1, v;
39     V<int> s;
38     PalTree() : n0(poolptr++), n1(poolptr++), v(n1) {
75         *n0 = {-1};
af         *n1 = {0, 1, n0};
b8     }
3f     np add_char(int c) {
b8         int i = s.size();
82         s.push_back(c);
bd         if (v->len == i) v = v->li;
34         while (s[i - v->len - 1] != c) v = v->li;
87         if (!v->nx[c]) {
a1             np n = poolptr++, m = v->li, l = n1;
14             if (v->len >= 0) {
b6                 while (s[i - m->len - 1] != s[i]) m = m->li;
a6                 l = m->nx[c];
4b             }
6f             v->nx[c] = n;
9c             *n = {v->len + 2, v->len + 2 - l->len, l, v};
96             n->sli = n->diff == l->diff ? l->sli : n;
da         }
95         return v = v->nx[c];
56     }
c3 };
```

rolling-hash.hpp

md5: 0d786b

```
c4 struct RollingHash {
c4     using ull = unsigned long long;
c7     constexpr static ull MOD = (1ULL << 61) - 1;
    // a < MOD, b < MOD必須
f5     constexpr static ull add(ull a, ull b) {
41         if ((a += b) >= MOD) a -= MOD;
32         return a;
ce     }
    // a < MOD, b < MOD必須
51     constexpr static ull mul(ull a, ull b) {
40         __uint128_t c = (__uint128_t)(a)*b;
7b         return add((ull)(c >> 61), (ull)(c & MOD));
21     }
83     void expand(int n) {
bd         while (int(power.size()) <= n)
2c             power.push_back(mul(power.back(), base));
05     }

b7     unsigned base;
08     vector<ull> power;
aa     RollingHash() {
69         base = random_device{}();
94         power = {1};
cf     }
22     vector<ull> build(const auto& s) {
bf         vector<ull> res;
d1         res.reserve(ssize(s) + 1);
7f         res.push_back(0);
f4         for (auto c : s) {
bc             res.push_back(add(mul(res.back(), base), c));
0e         }
03         return res;
1d     }
61     ull query(const vector<ull>& hash, int l, int r) {
2b         expand(r - l);
8f         return add(hash[r], MOD - mul(hash[l], power[r - l]));
ca     }
0d };
```

run_enum.hpp

md5: cea301

```
    // itereator バージョンの z_algorithm が必要
66 struct Run {
e5     int p, l, r;
19 }; // period, l, r
76 V<Run> EnumerateRuns(string S) {
80     int n = S.size();
16     V<Run> res;
```

```
91 auto f = [&](auto& f, int l, int r) -> void {
83     int len = r - l, lz = len / 2, rz = len - lz;
6c     if (len < 2) return;
d9     V<int> tg(len * 2);
50     REP(t, 2) REP(i, len) tg[t * len + i] = S[l + i];
1b     f(f, l + lz, r);
fb     f(f, l, l + lz);
10     auto a = z_algorithm(tg.begin() + lz, tg.end());
96     auto b = z_algorithm(tg.rbegin() + rz, tg.rend());
c1     auto ch = [&](int p, int lx, int rx) {
66         if (l != 0 && lx == lz) return;
6b         if (r != n && rx == rz) return;
1a         if (lx + rx < p * 2) return;
d7         res.push_back({p, l + lz - lx, l + lz + rx});
91     };
65     for (int p = 1; p <= lz; p++)
09         if (a[len - p] < p)
73             ch(p, min(lz, b[p] + p), min(rz, a[len - p]));
73     for (int p = 1; p <= rz; p++)
57         ch(p, min(lz, b[len - p]), min(rz, a[p] + p));
b5 };
be f(f, 0, n);
07 return res;
ce }
```

suffix_array.hpp

md5: 43ef37

```
62 V<ll> sa_is(auto s, ll B = 200) {
2d     ll n = s.size(), m = 0, b = 0;
4f     if (n <= 1) return {0};
c4     V<ll> a(n + 1, -1), sa(n), ls(n);
3b     REP(i, n) a[i] = s[i];
6d     for (ll i = n - 2; i >= 0; i--)
c8         ls[i] =
36             a[i] < a[i + 1] || a[i] == a[i + 1] && ls[i + 1];
76     auto induce = [&](const V<ll>& lms) {
ec         REP(i, n) sa[i] = -1;
4d         V<ll> zs(B + 1), zl(B + 2), yl(B + 2);
be         REP(i, n) ls[i] ? zl[a[i] + 1]++ : zs[a[i]]++;
fb         REP(i, B + 1) yl[i] = zl[i + 1] += zs[i] += zl[i];
fd         REP(i, m) sa[zs[a[lms[i]]]++] = lms[i];
2c         sa[zl[a[n - 1]]++] = n - 1;
61         for (ll v : sa)
5c             if (v-- > 0 && !ls[v]) sa[zl[a[v]]++] = v;
da         REP(i, n)
c8             if (ll v = sa[n - 1 - i]; v-- > 0 && ls[v])
4a                 sa[--yl[a[v]]] = v;
de     };
8a     V<ll> f(n + 1, -1), lms(n + 1, n);
de     REP(i, n)
75         if (i && !ls[i - 1] && ls[i]) lms[f[i] = m++] = i;
35     induce(lms);
82     if (m) {
32         V<ll> res, nx(m);
b7         REP(i, n) if (f[sa[i]] >= 0) res.push_back(sa[i]);
ef         REP(i, m - 1) {
1c             ll l = res[i], r = res[i + 1];
ed             ll el = lms[f[l] + 1], er = lms[f[r] + 1];
ec             while (l < el && r < er && a[l] == a[r]) l++, r++;
5b             nx[f[res[i + 1]]] = b += l < el || r < er;
57         }
93         auto q = sa_is(nx, b);
c8         for (ll& x : q) x = lms[x];
9d         induce(q);
90     }
b2     return sa;
43 }
```

suffix_array_doubling.hpp

md5: 073492

```
24 template <class Str>
4d V<ll> suffix_array(Str s) {
cf     ll n = s.size();
0d     V<ll> sa(n), rsa(n), tmp(n);
c1     REP(i, n) sa[i] = i;
20     REP(i, n) rsa[i] = s[i]; // 必要ならここで座圧
93     for (ll k = 1; k < n; k *= 2) {
0f         auto cmp = [&](ll x, ll y) {
8f             if (rsa[x] != rsa[y]) return rsa[x] < rsa[y];
c3             ll rx = x + k < n ? rsa[x + k] : -1;
f6             ll ry = y + k < n ? rsa[y + k] : -1;
99             return rx < ry;
31         };

```

```
3d         sort(sa.begin(), sa.end(), cmp);
2f         tmp[sa[0]] = 0;
e9         REP(i, n - 1)
bf             tmp[sa[i + 1]] = tmp[sa[i]] + cmp(sa[i], sa[i + 1]);
29         REP(i, n) rsa[sa[i]] = tmp[i];
63     }
bf     return sa;
07 }
```

suffix_automaton.hpp

md5: 0acc15

```
92 struct SuffixAutomaton {
d1     using Char = char;
cb     using Str = string;

    /*
        Node information

        構築時に作る情報を増やしたいなら、
        - Node の定義
        - MakeNode() での初期化
        - clone のときの deep copy
        を変更する

        len :
            マッチするsubstrのうちlongest (4)
    */
ad     struct Node {
4c         using NP = Node*;
19         int id;
66         int len;
62         NP link;
c9         map<Char, NP> next;

ec         Node() {}
c7         int getLongest() { return len; }
49         int getShortest() { return link ? link->len + 1 : 0; }
3e         void putNext(Char c, NP to) { next[c] = to; }
c1         bool hasNext(Char c) { return next.count(c); }
a9         NP getNext(Char c) { return hasNext(c) ? next[c] : 0; }
bc     };
c0     using NP = Node*;
15     NP last;
c0     int sz; // the number of nodes
d8     V<NP> nodes;
0e     SuffixAutomaton(int N) : sz(0), nodes(2 * N, 0) {
17         last = nodes[0] = MakeNode(0);
69     }

4c     NP MakeNode(int len) {
98         NP v = nodes[sz] = new Node();
7d         v->id = sz++;
db         v->len = len;
b2         v->link = 0;
f8         return v;
3d     }
    /*
        Add c to nodes[last]
        return the new node id
    */
e5     void add_char(Char c) {
77         NP p = last;
ad         last = MakeNode(last->len + 1);
ca         for (; p && !p->hasNext(c); p = p->link)
97             p->next[c] = last;
19         if (!p) {
06             last->link = nodes[0];
7a         } else {
62             NP q = p->getNext(c);
60             if (p->len + 1 == q->len) {
cd                 last->link = q;
64                 return;
87             }
            // clone!
c9             NP w = MakeNode(p->len + 1);
            /*
                deep copy !
            */
8c             w->next = q->next;
de             w->link = q->link;
e3             for (; p && p->getNext(c) == q; p = p->link)
16                 p->next[c] = w;
f9             q->link = last->link = w;

```

```
5f     }
75   }
43   void TopologicalSort() {
9a     V<int> deg(sz);
63     V<NP> ord(sz);
c5     REP(v, sz)
c1       for (auto it : nodes[v]->next) deg[it.second->id]++;
1e     ord[0] = nodes[0];
6e     int idx = 1;
b8     for (NP n : ord)
65       for (auto it : nodes[n->id]->next)
85         if (!--deg[it.second->id]) ord[idx++] = it.second;
15     swap(nodes, ord);
1c     REP(i, sz) nodes[i]->id = i;
24   }
0a   };
```

geometry

primitive.hpp

md5: 8fcbf7

```
7f using D = double;
9b const D PI = acos(D(-1)), EPS = 1e-10;
e0 int sgn(D a) { return (a < -EPS) ? -1 : (a > EPS); }
ac int sgn(D a, D b) { return sgn(a - b); }
a2 struct P {
56   D x, y;
d9   P(D x = D(), D y = D()) : x(x), y(y) {}
83   P operator+(P r) const { return {x + r.x, y + r.y}; }
f7   P operator-(P r) const { return {x - r.x, y - r.y}; }
8b   P operator*(P r) const {
2e     return {x * r.x - y * r.y, x * r.y + y * r.x};
35   }
90   P operator*(D r) const { return {x * r, y * r}; }
0e   P operator/(D r) const { return {x / r, y / r}; }
ff   P& operator+=(P r) { return *this = *this + r; }
c2   P& operator-=(P r) { return *this = *this - r; }
97   P& operator*=(P r) { return *this = *this * r; }
e2   P& operator*=(D r) { return *this = *this * r; }
69   P& operator/=(D r) { return *this = *this / r; }
7c   P operator-() const { return {-x, -y}; }
8c   bool operator<(P r) const {
94     return 2 * sgn(x, r.x) + sgn(y, r.y) < 0;
34   }
cb   bool operator==(P r) const {
cf     return sgn((*this - r).rabs()) == 0;
ac   }
4b   bool operator!=(P r) const { return !(*this == r); }
b9   D norm() const { return x * x + y * y; }
f0   D abs() const { return sqrt(norm()); }
8f   D rabs() const {
8f     return max(std::abs(x), std::abs(y));
18   }
// robust abs
19   D arg() const { return atan2(y, x); }
76   pair<D, D> to_pair() const { return {x, y}; }
da   static P polar(D le, D th) {
4d     return {le * cos(th), le * sin(th)};
b4   }
30 };
46 ostream& operator<<(ostream& os, const P& p) {
d9   return os << "P(" << p.x << ", " << p.y << ")";
d2 }
6d struct L {
67   P s, t;
9c   L(P s = P(), P t = P()) : s(s), t(t) {}
03   P vec() const { return t - s; }
54   D abs() const { return vec().abs(); }
2b   D arg() const { return vec().arg(); }
4b };
c3 ostream& operator<<(ostream& os, const L& l) {
be   return os << "L(" << l.s << ", " << l.t << ")";
fa }
7d D crs(P a, P b) { return a.x * b.y - a.y * b.x; }
28 D dot(P a, P b) { return a.x * b.x + a.y * b.y; }
// cross(a, b) is too small?
19 int sgncrs(P a, P b) {
de   D cr = crs(a, b);
65   if (abs(cr) <= (a.rabs() + b.rabs()) * EPS) return 0;
19   return (cr < 0) ? -1 : 1;
b3 }
// -2, -1, 0, 1, 2 : front, clock, on, cclock, back
```

```
f1 int ccw(P b, P c) {
e8   int s = sgncrs(b, c);
d4   if (s) return s;
25   if (!sgn(c.rabs()) || !sgn((c - b).rabs())) return 0;
32   if (dot(b, c) < 0) return 2;
c7   if (dot(-b, c - b) < 0) return -2;
c0   return 0;
b5 }
12 int ccw(P a, P b, P c) { return ccw(b - a, c - a); }
8f int ccw(L l, P p) { return ccw(l.s, l.t, p); }
```

intersect.hpp

md5: 899d98

```
cb P project(L l, P p) {
b3   P v = l.vec();
b5   return l.s + v * dot(v, p - l.s) / v.norm();
a4 }
79 bool insSL(L s, L l) {
76   int a = ccw(l, s.s), b = ccw(l, s.t);
6a   return a % 2 == 0 || b % 2 == 0 || a != b;
e4 }
34 bool insSS(L s, L t) {
3b   int a = ccw(s, t.s), b = ccw(s, t.t);
4b   int c = ccw(t, s.s), d = ccw(t, s.t);
60   return a * b <= 0 && c * d <= 0;
8d }
cd D distLP(L l, P p) {
ee   return abs(crs(l.vec(), p - l.s)) / l.abs();
69 }
42 D distSP(L s, P p) {
ae   P q = project(s, p);
40   if (ccw(s, q) == 0) return (p - q).abs();
18   return min((s.s - p).abs(), (s.t - p).abs());
e5 }
01 D distSS(L s, L t) {
03   if (insSS(s, t)) return 0;
40   return min({distSP(s, t.s), distSP(s, t.t),
47     distSP(t, s.s), distSP(t, s.t)});
89 }
```

cross_point.hpp

md5: a16c01

crossLL,crossSSの返り値は、1:1点を共有、0:共有点なし、-1:無数の共有点(重なり)

```
d7 int crossLL(L l, L m, P& r) {
fa   D cr1 = crs(l.vec(), m.vec()),
27   cr2 = crs(l.vec(), l.t - m.s);
c9   if (sgncrs(l.vec(), m.vec()) == 0) {
16     r = l.s;
63     return sgncrs(l.vec(), l.t - m.s) ? 0 : -1;
0c   }
78   r = m.s + m.vec() * cr2 / cr1;
79   return 1;
ef }
fa int crossSS(L l, L m, P& r) {
f3   int u = crossLL(l, m, r);
1d   if (u == 0) return 0;
f9   if (u == -1) {
e9     r = max(min(l.s, l.t), min(m.s, m.t));
29     P q = min(max(l.s, l.t), max(m.s, m.t));
33     return (q < r) ? 0 : (q == r ? 1 : -1);
0e   }
a5   return ccw(l, r) == 0 && ccw(m, r) == 0 ? 1 : 0;
a1 }
```

polygon.hpp

md5: 78e152

insPolLは、多角形と直線の交点。sからtに向かう方向に並んでいる。直線からみると、多角形の内部が(すぐ左に見えるか)、(すぐ右に見えるか)の2つのプロパティがあり、それが切り替わる点一覧が返される。次の領域でのプロパティがsecondに2bitで表現される

```
6e using Pol = V<P>;
a6 D area2(const Pol& pol) {
34   D u = 0;
48   if (!pol.size()) return u;
b9   P a = pol.back();
39   for (auto b : pol) u += crs(a, b), a = b;
5f   return u;
0e }
```

```
// 0: outside, 1: on line, 2: inside
51 int contains(const Pol& pol, P p) {
4e   if (!pol.size()) return 0;
8d   int in = -1;
c8   P _a, _b = pol.back();
ba   for (auto now : pol) {
b1       _a = _b;
d7       _b = now;
c9       P a = _a, b = _b;
be       if (ccw(a, b, p) == 0) return 1;
78       if (a.y > b.y) swap(a, b);
3f       if (!(a.y <= p.y && p.y < b.y)) continue;
5a       if (sgn(a.y, p.y) ? (crs(a - p, b - p) > 0)
9c           : (a.x > p.x))
b7           in *= -1;
3b   }
cd   return in + 1;
b6 }

// 以下 cross_point.hpp が必要

06 Pol convex_cut(const Pol& po, L l) {
23   if (!po.size()) return Pol{};
af   Pol q;
b3   P a, b = po.back();
79   for (auto now : po) {
af       a = b;
8c       b = now;
9f       if (((ccw(l, a) % 2) * (ccw(l, b) % 2) < 0) {
70           P buf;
c2           crossLL(l, L(a, b), buf);
f8           q.push_back(buf);
e5       }
26       if (ccw(l, b) != -1) q.push_back(b);
e7   }
4c   return q;
a4 }

// (1:left) | (2: right) is inside between v[i] ~ v[i + 1]
ce V<pair<P, int>> insPoll(Pol pol, L l) {
c7   using Pi = pair<P, int>;
de   V<Pi> v;
23   P a, b = pol.back();
ac   for (auto now : pol) {
a8       a = b;
67       b = now;
5e       P p;
41       if (crossLL({a, b}, l, p) != 1) continue;
ce       int sa = ccw(l, a) % 2, sb = ccw(l, b) % 2;
00       if (sa > sb) swap(sa, sb);
be       if (sa != 1 && sb == 1) v.push_back({p, 1});
94       if (sa == -1 && sb != -1) v.push_back({p, 2});
69   }
4a   sort(v.begin(), v.end(), [&](Pi x, Pi y) {
31       return dot(l.vec(), x.first - l.s) <
79           dot(l.vec(), y.first - l.s);
9f   });
8d   int m = int(v.size());
4b   V<Pi> res;
d6   for (int i = 0; i < m; i++) {
08       if (i) v[i].second ^= v[i - 1].second;
b1       if (!res.empty() && res.back().first == v[i].first)
bf           res.pop_back();
29       res.push_back(v[i]);
04   }
4b   return res;
78 }
```

circle.hpp

md5: 504af5

```
39 struct C {
b4   P p;
86   D r;
6a   C(P p = P(), D r = D()) : p(p), r(r) {}
aa };

// need Intersect/distLP, r.sはよりl.sに近い
a3 int crossCL(C c, L l, L& r) {
54   D u = distLP(l, c.p);
6d   int si = sgn(u, c.r);
17   if (si == 1) return 0;
87   P v = project(l, c.p);
27   P di =
6b       (si == 0)
04       ? P(0, 0)
```

```
77       : l.vec() * (sqrt(c.r * c.r - u * u) / l.abs());
ad   r = L(v - di, v + di);
7d   if (si == 0) return 1;
9c   return 2;
de }

// need Intersect/distLP, r.sはよりl.sに近い
d6 int crossCS(C c, L s, L& l) {
0b   if (!crossCL(c, s, l)) return 0;
7b   bool f1 = !ccw(s, l.s), f2 = !ccw(s, l.t);
2d   if (f1 && f2) return 2;
83   if (!f1 && !f2) return 0;
94   if (f1)
41       l.t = l.s;
48   else
11       l.s = l.t;
fb   return 1;
5c }

// return number of cross point
// l, rはcから見た交点の角度、[l, r]がdに覆われている
c2 int crossCC(C c, C d, D& l, D& r) {
97   if (c.p == d.p) {
90       assert(sgn(c.r - d.r));
// prohibit same circles
return 0;
15   }
de   }
1c   D di = (c.p - d.p).abs(), bth = (d.p - c.p).arg();
87   D costh =
52       (c.r * c.r + di * di - d.r * d.r) / (2 * c.r * di);
d6   int ty =
5c       min(sgn(c.r + d.r, di), sgn(di, abs(c.r - d.r)));
8d   if (ty == -1) return 0;
ad   if (ty == 0) costh = sgn(costh);
c7   D th = acos(max(D(-1), min(D(1), costh)));
88   l = bth - th;
cb   r = bth + th;
0d   return ty + 1;
8a }

// C(P(0, 0), r)とTri((0, 0), a, b)の共有面積
9e D area2CT(C c, P _a, P _b) {
fc   P a = _a - c.p, b = _b - c.p;
03   D r = c.r;
99   if (a == b) return 0;
99   auto single = [&](P x, P y, bool tri) {
94       if (tri) return crs(x, y);
95       return r * r * ((y * P(x.x, -x.y)).arg());
09   };
02   bool ia = sgn(a.abs(), r) != 1,
92       ib = sgn(b.abs(), r) != 1;
8c   if (ia && ib) return single(a, b, true);
20   L l;
43   if (!crossCS(C(P(0, 0), r), L(a, b), l))
50       return single(a, b, false);
59   return single(a, l.s, ia) + single(l.s, l.t, true) +
94       single(l.t, b, ib);
bf }

// p, cの共有面積
05 D area2CPol(C c, const V<P>& po) {
f8   D sm = 0;
b4   P a, b = po.back();
17   for (auto p : po) {
88       a = b;
c1       b = p;
0e       sm += area2CT(c, a, b);
c9   }
5b   return sm;
50 }
```

circle_tangent.hpp

md5: abf1ee

```
// pからcに接線を引く、交点はp1, p2
2a int tangent(C c, P p, P& p1, P& p2) {
21   D di = (c.p - p).abs();
dc   int si = sgn(c.r, di);
1d   if (si == 1) return 0;
cb   D th = acos(min(D(1), c.r / di));
e5   D ar = (p - c.p).arg();
36   p1 = c.p + P::polar(c.r, ar - th);
f5   p2 = c.p + P::polar(c.r, ar + th);
b8   if (si == 0) return 1;
06   return 2;
b7 }

// 共通接線
c7 int tangent(C c, C d, L& l, L& r, bool inter) {
```

```
30 D di = (d.p - c.p).abs(), ar = (d.p - c.p).arg();
c8 if (sgn(di) == 0) {
dd     assert(sgn(c.r - d.r));
        // prohibit same circles
09     return 0;
76 }
aa D costh = c.r + (inter ? d.r : -d.r);
3a int si = sgn(abs(costh), di);
a4 costh /= di;
74 if (si == 1) return 0;
6e if (si == 0) costh = sgn(costh);
f4 D th = acos(max(D(-1), min(D(1), costh)));
52 P base;
7a base = P::polar(1, ar - th);
e7 l = L(c.p + base * c.r,
bf     d.p + base * d.r * (inter ? -1 : 1));
1a base = P::polar(1, ar + th);
r = L(c.p + base * c.r,
64     d.p + base * d.r * (inter ? -1 : 1));
63 if (si == 0) return 1;
6f return 2;
ab }
```

convex_hull.hpp

md5: 6bb478

```
7a C circum_circle(P a, P b, P c) {
2b b -= a;
a8 c -= a;
da D s = 2 * crs(b, c);
3b P x = {b.x, c.x};
08 P y = {b.y, c.y};
3f P z = {b.norm(), c.norm()};
67 P r = P{crs(z, y), crs(x, z)} / s;
ec return C(r + a, r.abs());
85 }
0a C smallest_circle(V<P>& p, int i, array<P, 3> q, int j) {
78 if (i == int(p.size())) {
c7 if (j == 0) return C(P(0, 0), -1);
65 if (j == 1) return C(q[0], 0);
d8 if (j == 2)
3f     return C((q[0] + q[1]) / D(2.0),
f5             (q[0] - q[1]).abs() / D(2.0));
52 if (j == 3) return circum_circle(q[0], q[1], q[2]);
19 assert(false);
d0 }
57 C c = smallest_circle(p, i + 1, q, j);
74 if (sgn((p[i] - c.p).abs(), c.r) == 1) {
7a q[j] = p[i];
79 return smallest_circle(p, i + 1, q, j + 1);
4c }
06 return c;
b9 }
3b C smallest_circle(V<P> p) {
14 random_shuffle(begin(p), end(p));
f1 return smallest_circle(p, 0, array<P, 3>(), 0);
3d }
```

convex_hull.hpp

md5: 6bb478

```
// p must be sorted and uniqued!
16 V<P> convex_down(V<P> ps, bool on_edge = 0) {
75 assert(ps.size() >= 2);
dc int r = -1, f = !on_edge;
f3 for (P d : ps) {
40 while (r > 0 && ccw(ps[r - 1], ps[r], d) % 2 < f) r--;
ea ps[++r] = d;
ae }
f4 ps.resize(r + 1);
71 return ps;
76 }
// counter-clockwise
d9 V<P> convex(V<P> ps, bool on_edge = 0) {
23 sort(ps.begin(), ps.end());
42 ps.erase(unique(ps.begin(), ps.end()), ps.end());
6f if (ps.size() <= 1) return ps;
e1 auto dw = convex_down(ps, on_edge);
ed reverse(ps.begin(), ps.end());
3c auto up = convex_down(ps, on_edge);
a7 dw.insert(dw.begin(), up.begin() + 1, up.end() - 1);
ee return dw;
6b }
```

convex_polygon_diameter.hpp

md5: 3fa6ae

```
2e D diameter(const V<P>& p) {
ef int n = int(p.size());
b8 if (n == 2) return (p[1] - p[0]).abs();
2a int x = 0, y = 0;
3c for (int i = 1; i < n; i++) {
4a if (p[i] < p[x]) x = i;
4e if (p[i] < p[y]) y = i;
a2 }
3b D ans = 0;
63 int sx = x, sy = y;
/*
do {
...
} while (sx != x || sy != y);
で1周する
*/
ad while (sx != y || sy != x) {
76 ans = max(ans, (p[x] - p[y]).abs());
ba int nx = (x + 1 < n) ? x + 1 : 0,
0f ny = (y + 1 < n) ? y + 1 : 0;
23 if (crs(p[nx] - p[x], p[ny] - p[y]) < 0)
93 x = nx;
da else
99 y = ny;
85 }
eb return ans;
3f }
```

closest_pair.hpp

md5: 0f5400

```
9e tuple<D, int, int> cp_rec(vector<pair<P, int>>& psi, int l,
41 int r) {
39 if (l + 1 == r) return {1e100, -1, -1};
28 int m = midpoint(l, r);
12 D x = psi[m].first.x;
8c auto dd = min(cp_rec(psi, l, m), cp_rec(psi, m, r));
33 auto [d, di, dj] = dd;
ba inplace_merge(psi.begin() + l, psi.begin() + m,
95 psi.begin() + r, [&](auto a, auto b) {
cf return a.first.y < b.first.y;
59 });
1e vector<int> pp;
6c for (int a = l; a < r; a++) {
80 auto [p, i] = psi[a];
cd if (p.x <= x - d || x + d <= p.x) continue;
f4 for (int b : pp | views::reverse) {
7d auto [q, j] = psi[b];
2e D dx = p.x - q.x, dy = p.y - q.y;
33 if (dy >= d) break;
08 D nd = sqrt(dx * dx + dy * dy);
4c if (d > nd) d = nd, di = i, dj = j;
e2 }
73 pp.push_back(a);
89 }
30 return {d, di, dj};
ee }
38 tuple<D, int, int> closest_pair(const vector<P>& ps) {
b2 int n = ssize(ps);
f3 vector<pair<P, int>> psi(n);
7a REP(i, n) psi[i] = {ps[i], i};
93 sort(psi.begin(), psi.end(), [](auto a, auto b) {
62 return a.first.x < b.first.x;
fb });
24 return cp_rec(psi, 0, n);
0f }
```

halfplane_intersection.hpp

md5: c4d3af

```
34 int argcmp(P l, P r) {
9d auto psgn = [&](P p) {
48 if (int u = sgn(p.y)) return u;
f5 if (sgn(p.x) == -1) return 2;
a9 return 0;
de };
86 int lsgn = psgn(l), rsgn = psgn(r);
de if (lsgn < rsgn) return -1;
00 if (lsgn > rsgn) return 1;
96 return sgn(r, l);
4d }
// line = (s,t) としたとき、 s-->t
```

```
// の方向に見たときの右側を切り捨てる
78 V<L> halfplane_intersection(V<L> lines) {
ea   auto st = move(lines);
1b   sort(st.begin(), st.end(), [](L a, L b) {
bd       if (int u = argcmp(a.vec(), b.vec())) return u == -1;
d4       return sgnocrs(a.vec(), b.s - a.s) < 0;
37   });
1c   st.erase(unique(st.begin(), st.end(),
ee       [](L a, L b) {
41           return !argcmp(a.vec(), b.vec());
57       }),
e6       st.end());
38   ll l = 0, r = -1, err = 0;
3c   for (auto& g : st) {
6f       auto need = [&](L a, L b, L c) {
ad           D ab_dw = crs(a.vec(), b.vec()),
67           ab_up = crs(a.vec(), a.t - b.s);
eb           D bc_dw = crs(b.vec(), c.vec()),
81           bc_up = crs(c.t - b.s, c.vec());
18           if (ab_dw <= 0 || bc_dw <= 0) return true;
54           bool f = bc_up * ab_dw > bc_dw * ab_up;
67           if (!f && crs(a.vec(), c.vec()) < 0) err = 1;
5e           return f;
39       };
88       while (r - l >= 1 && !need(g, st[l], st[l + 1])) l++;
c7       while (r - l >= 1 && !need(st[r - 1], st[r], g)) r--;
68       if (r - l < 1 || need(st[r], g, st[l])) st[++r] = g;
99       if (err) return {};
35   }
72   if (r - l == 1 && !sgnocrs(st[l].vec(), st[r].vec()) &&
9f       sgnocrs(st[l].vec(), st[r].s - st[l].s) <= 0)
f5       return {};
22   return V<L>(st.begin() + l, st.begin() + r + 1);
c4 }
```

segment_ordering.hpp

md5: 05fd10

```
68 struct LI {
0d   L l;
40   ll i;
62   L reg() const { return l.s < l.t ? l : L{l.t, l.s}; }
5d   static ll d(P a, P b, P c) {
2a       return c < a || b < c ? 0 : sgnocrs(a, b, c);
a3   }
8d   bool operator<(LI b) const {
c0       L l = reg(), r = b.reg();
3e       if (l.t < r.s || r.t < l.s) return 0;
18       ll f = 0;
52       f += d(l.s, l.t, r.s) - d(r.s, r.t, l.s);
88       f += d(l.s, l.t, r.t) - d(r.s, r.t, l.t);
30       if (f) return 0 < f;
af       return !(r.s < l.t) && (l.s < r.t || i < b.i);
d1   }
84   };
e0 V<ll> orderSegments(V<L> e, ll inf = 1ll << 30) {
0e   ll n = e.size(), f = 0;
33   for (auto& l : e)
23       if (l.t < l.s) swap(l.s, l.t);
1d   V<pair<P, int>> X(n * 2);
28   REP(i, n) X[i] = {e[i].s, i};
45   REP(i, n) X[n + i] = {e[i].t, n + i};
69   V<ll> d(n), res(n), ord(n, -1);
dc   V<V<ll>> adj(n);
67   std::set<LI> ds;
42   V<std::set<LI>::iterator> its(n, ds.end());
b6   ds.insert({{-inf, -inf}, {inf, -inf}}, -1);
25   ds.insert({{-inf, inf}, {inf, inf}}, -1);
f2   stable_sort(X.begin(), X.end());
5a   auto addEdge = [&](ll l, ll r) {
43       if (l >= 0 && r >= 0) adj[l].push_back(r), d[r]++;
fe   };
18   for (auto [x, i] : X) {
34       if (i < n) {
17           its[i] = ds.insert({e[i], i}).first;
11           auto l = its[i], r = its[i];
91           addEdge((--l)->i, i), addEdge(i, (++r)->i);
74       } else {
3a           i -= n;
7e           auto l = its[i], r = its[i];
45           addEdge((--l)->i, (++r)->i);
74           ds.erase(its[i]);
1f       }
7c   }
```

```
c5   REP(i, n) if (!d[i]) ord[f++] = i;
c8   for (ll v : ord)
7d       if (v >= 0)
f8           for (ll w : adj[v])
10               if (!--d[w]) ord[f++] = w;
e7   REP(i, n) res[ord[i]] = i;
47   return res;
05 }
```

other

hook_length_formula.hpp

md5: cf7510

```
1e Mint Hook_Length_Formula(vector<int>& A) {
97   assert(is_sorted(A.begin(), A.end()));
36   if (A.empty()) return Mint(1);
cc   vector<int> L(A.back(), 0);
ec   Mint n(1), d(1);
61   int s = 0;
bf   for (int l : A) REP(j, l) {
15       n *= Mint(++s);
f7       d *= Mint(L[j]++ + l - j);
f1   }
25   return n / d;
cf }
```

matrix.hpp

md5: a555f8

```
12 const Mint zero(0);
c7 const Mint one(1);
f9 bool iszero(Mint x) { return x.v == 0; }
20 template <class T>
03 struct Matrix {
0a   int H, W;
f6   V<V<T>> a;
81   Matrix(V<V<T>> v)
d8       : H(v.size()), W(v[0].size()), a(move(v)) {}
1c   Matrix(int H = 0, int W = 0)
a6       : H(H), W(W), a(H, V<T>(W)) {}
b7   static Matrix E(int n) {
c1       Matrix a(n, n);
20       REP(i, n) a.set(i, i, 1);
e5       return a;
1b   }
05   T at(int i, int j) const { return a[i][j]; }
7d   void set(int i, int j, T v) { a[i][j] = v; }
/*
左から var 列が階段行列になるように、行基本変形を施す。
(e.g. 逆行列 は右に単位行列おいてから掃き出す)
rank,determinant を返す。 in placeに更新されることに注意
*/
36   pair<int, T> sweep(int var) {
44       int i = 0, rank = 0;
60       T d = one;
3d       REP(j, var) {
12           for (i = rank; i < H && iszero(a[i][j]);) i++;
4a           if (i == H) continue;
31           if (i != rank) d = -d;
0e           swap(a[i], a[rank]);
3e           i = rank++;
24           T t = a[i][j];
eb           REP(k, W) a[i][k] /= t;
c6           d *= t;
d5           REP(k, H) if (k != i) {
fa               t = a[k][j];
90               REP(l, W) a[k][l] -= a[i][l] * t;
32           }
e6       }
e3       if (H != W || rank != H) d *= zero;
67       return {rank, d};
a7   }
2b   Matrix tr() {
df       auto B = Matrix(W, H);
da       REP(i, H) REP(j, W) B.set(j, i, at(i, j));
37       return B;
49   }
76   int getrank() {
d3       auto B = H < W ? *this : tr();
cb       return B.sweep(B.W).first;
b6   }
52   T getdet() { return Matrix(*this).sweep(W).second; }
12   Matrix inv() {
```

```
3a      assert(H == W);
6f      int N = H;
a1      Matrix X(N, 2 * N), B(N, N);
89      REP(i, N) REP(j, N) X.set(i, j, a[i][j]);
99      REP(i, N) X.set(i, i + N, one);
45      if (X.sweep(N).first < N) return Matrix();
e4      REP(i, N) REP(j, N) B.set(i, j, X.at(i, j + N));
6d      return B;
ab    }
80    friend ostream& operator<<(ostream& o, const Matrix& A) {
79      REP(i, A.H) {
99        REP(j, A.W) o << A.a[i][j] << " ";
3a        o << endl;
6a      }
3b      return o;
61    }
a5  };
```

surreal.hpp md5: 266388

```
27 struct Surreal {
4c   using S = Surreal;
ac   using Int = long long;
13   Int p, q; // val = p / 2**q
e2   Surreal(Int p = 0, Int q = 0) : p(p), q(q) {
f1     while (p % 2 == 0 && q) p /= 2, q--;
f8   }
39   S operator+(S r) const {
fe     auto c = max(q, r.q);
f4     return S((p << (c - q)) + (r.p << (c - r.q)), c);
eb   }
5d   S operator-() const { return {-p, q}; }
0c   bool operator<(S r) const { return (r + -*this).p > 0; }
2f   S lc() const {
99     if (!q && p <= 0) return p - 1;
31     return {p * 2 - 1, q + 1};
fb   }
65   friend S operator|(S l, S r) {
88     assert(l < r);
ab     S v = 0;
bf     while (!(l < v && v < r))
f1       v = l < v ? v.lc() : -(-v).lc();
2e     return v;
e2   }
26 };
```

memo

Nimber.md

$$[(ab)_h, (ab)_l] = [(a_h \oplus a_l)(b_h \oplus b_l) \oplus a_l b_l, a_h b_h 2^{2^d-1} \oplus a_l b_l]$$

$$[(\text{inv } x)_h, (\text{inv } x)_l] = \text{inv}((x_h \oplus x_l)x_l \oplus x_h x_h 2^{2^d-1})[x_h, x_h \oplus x_l]$$

ラグランジュの反転公式.md

F, G が多項式で $F(G(x)) = G(F(x)) = x$ のとき

$$[x^n]G(x)^k = \frac{k}{n}[x^{-k}]F(x)^{-n}$$

$$[x^n]H(G(x)) = \frac{1}{n}[x^{-1}]H'(x)F(x)^{-n}$$

二部グラフ.md

- 最小点被覆 : L のマッチされていない頂点から、到達不能な L の頂点と到達可能な R の頂点。
- 最大独立点集合 : 最小点被覆の補集合
- 最小辺被覆 : 最大マッチングに加えて、マッチングで覆われていない各頂点に接続する辺を1つずつ選ぶ。

五心.md

$$f(w_a, w_b, w_c) = (w_a \vec{a} + w_b \vec{b} + w_c \vec{c}) / (w_a + w_b + w_c) \text{ として、}$$

- 外心 $\vec{o} = f(\sin 2A, \sin 2B, \sin 2C)$
- 内心 $\vec{i} = f(|\vec{a}|, |\vec{b}|, |\vec{c}|)$ (引数の符号を1個変えると傍心)
- 垂心 $\vec{h} = f(\tan A, \tan B, \tan C)$
- 重心 $\vec{g} = f(1, 1, 1) = (2\vec{o} + \vec{h})/3$

数え上げ公式.md

オイラーグラフについて、 Eulerian Cycle の個数は $C_v \times \prod_{w \in V} (\text{outdeg}_w - 1)!$ (C_v は v を根とする有向全域木の個数、 v に依存しない)

v に向かう有向全域木の個数は、次の行列の v 行目と v 列目を除いた行列の行列式に等しい：辺 $u \rightarrow v$ に対して $a_{uv} = 1, a_{uu} = 1$.

整数.md

二項係数

$$\binom{n}{k} = \binom{n-1}{k-1} + \binom{n-1}{k}$$

$$\binom{n}{k} = \frac{n}{k} \binom{n-1}{k-1}$$

$$\binom{L}{k} + \cdots + \binom{R-1}{k} = \binom{R}{k+1} - \binom{L}{k+1}$$

第一種スターリング数

$c(n, k)$: $1, 2, \dots, n$ の順列で巡回置換 k 個に分割できるものの個数

$$c(n, k) = c(n-1, k-1) + (n-1)c(n-1, k)$$

$$\text{母関数 (k)}: \sum_{k=0}^N x^k c(N, k) = \prod_{i=0}^{N-1} (x - i)$$

$$\text{指数型母関数 (n)}: \sum_{n=0}^{\infty} (x^n/n!) c(n, K) = (\log(1+x))^K / K!$$

第二種スターリング数

$S(n, k)$: $1, 2, \dots, n$ を k 個の区別しない集合に分割する方法の数

$$S(n, k) = S(n-1, k-1) + kS(n-1, k)$$

$$x^n = \sum_{k=0}^n S(n, k) x(x-1) \cdots (x-k+1)$$

$$\text{指数型母関数 (n)}: \sum_{n=0}^{\infty} (x^n/n!) S(n, K) = (e^x - 1)^K / K!$$

$$\text{母関数 (k)}: \sum_{k=0}^N x^k S(N, k) = \left(\sum_{i=0}^{\infty} (-1)^i x^i / i!\right) \left(\sum_{i=0}^{\infty} i^N x^i / i!\right)$$

ベル数

B_n : $1, 2, \dots, n$ をいくつかの集合に分割する方法の数

n	0	1	2	3	4	5	6	7	8	9	10
B_n	1	1	2	5	15	52	203	877	4140	21147	115975

$$B_{n+1} = \sum_{k=0}^n \binom{n}{k} B_k$$

$$B_n = e^{-1} \sum_{k=0}^{\infty} k^n / k!$$

$$\text{指数型母関数}: \exp(\exp x - 1) = \sum_{n=0}^{\infty} B_n (x^n/n!)$$

カタラン数

C_n : n 個の (と) を括弧列になるように並べる方法の数

n	0	1	2	3	4	5	6	7	8	9	10
C_n	1	1	2	5	14	42	132	429	1430	4862	16796

$$C_n = \frac{1}{n+1} \binom{2n}{n} = \frac{(2n)!}{(n+1)n!}$$

$$C_{n+1} = \frac{2(2n+1)}{n+2} C_n = \sum_{k=0}^n C_k C_{n-k}$$

$$\text{母関数の乗乗}: C(x)^k = ((1 - \sqrt{1 - 4x})/2x)^k = \sum_{n=0}^{\infty} \frac{k}{n+k} \binom{2n+k-1}{n} x^n$$

モンモール数

a_n : $1, 2, \dots, n$ の順列 P で $P_i \neq i$ となるものの個数

<i>n</i>	0	1	2	3	4	5	6	7	8	9	10
<i>a</i> _{<i>n</i>}		0	1	2	9	44	265	1854	14833	133496	1334961

$a_n = na_{n-1} + (-1)^n$

指数型母関数 : $\sum_{n=0}^{\infty} a_n x^n / n! = e^{-x} / (1 - x)$

分割数

P_n : n を正の整数の和として表す方法の数

<i>n</i>	0	1	2	3	4	5	6	7	8	9	10		20	30	40	50
P_n	1	1	2	3	5	7	11	15	22	30	42		627	5604	37338	204226

母関数 : $(\sum_{n=-\infty}^{\infty} (-1)^n x^{n(3n+1)/2})^{-1}$

ベルヌーイ数

指数型母関数 : $\sum_{0 \leq n} (x^n / n!) B_n = x / (e^x - 1)$

$1^K + 2^K + \cdots + N^K = K! \cdot \sum_{j=0}^K (B_j / j!) (N^{K-j+1} / (K - j + 1)!)$

母関数.md

$e^x = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \cdots = \sum_{n=0}^{\infty} \frac{x^n}{n!}$

$\log(1 + x) = x - \frac{x^2}{2} + \frac{x^3}{3} - \frac{x^4}{4} + \cdots = \sum_{n=1}^{\infty} \frac{(-1)^{n-1}}{n} x^n$

$\sqrt{1-x} = 1 - \frac{1}{2}x - \frac{1}{8}x^2 - \frac{1}{16}x^3 - \frac{5}{128}x^4 - \cdots = 1 - \sum_{n=1}^{\infty} \frac{(2n-2)!}{2^{2n-1}n!(n-1)!} x^n$

$\frac{1}{\sqrt{1-x}} = 1 + \frac{1}{2}x + \frac{3}{8}x^2 + \frac{5}{16}x^3 + \frac{35}{128}x^4 - \cdots = \sum_{n=0}^{\infty} \frac{(2n)!}{4^n n!} x^n$

燃やす埋める.md

条件	処理
<i>i</i> が 0 のとき <i>c</i> 失う	make_edge(i, G, c)
<i>i</i> が 1 のとき <i>c</i> 失う	make_edge(S, i, c)
<i>i</i> が 0 のとき <i>c</i> 得る	先に <i>c</i> を得る make_edge(S, i, c)

条件	処理
<i>i</i> が 1 のとき <i>c</i> 得る	先に <i>c</i> を得る make_edge(i, G, c)
<i>i</i> が 0, <i>j</i> が 1 のとき <i>c</i> 失う	make_edge(i, j, c)
<i>i</i> が 1, <i>j</i> が 0 のとき <i>c</i> 失う	make_edge(j, i, c)
<i>i</i> と <i>j</i> が異なるとき <i>c</i> 失う	make_edge(i, j, c) make_edge(j, i, c)
<i>i</i> が 0, <i>j</i> が 0 のとき <i>c</i> 得る	先に <i>c</i> を得る make_edge(S, k, c) make_edge(k, i, inf) make_edge(k, j, inf)
<i>i</i> が 1, <i>j</i> が 1 のとき <i>c</i> 得る	先に <i>c</i> を得る make_edge(i, k, inf) make_edge(j, k, inf) make_edge(k, G, c)

素数.md

素数の個数

<i>n</i>	10 ²	10 ³	10 ⁴	10 ⁵	10 ⁶	10 ⁷	10 ⁸	10 ⁹	10 ¹⁰
$\pi(n)$	25	168	1229	9592	78498	664579	5.76e+6	5.08e+7	4.55e+8

高度合成数

$\leq n$	10^3	10^4	10^5	10^6	10^7	10^8	10^9		
x	840	7560	83160	720720	8648640	73513440	735134400		
$d^0(x)$	32	64	128	240	448	768	1344		
$\leq n$	10^{10}	10^{11}	10^{12}	10^{13}	10^{14}	10^{15}	10^{16}	10^{17}	10^{18}
$d^0(x)$	2304	4032	6720	10752	17280	26880	41472	64512	103680

素数階乗

<i>n</i>	2	3	5	7	11	13	17	19	23	29
<i>n</i> #	2	6	30	210	2310	30030	510510	9.70e+6	2.23e+8	6.47e+9

階乗

4!	5!	6!	7!	8!	9!	10!	11!	12!	13!
24	120	720	5040	40320	362880	3.63e+6	3.99e+7	4.79e+8	6.23e+9